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Large contribution of light-dependent oxygen uptake to global O_2 cycling

by

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Dedication

To my grandmother, Isabel Diego, who brought me back to God and pushed me to strive for perfection in both the corporeal and the spiritual life.

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Abstract

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The oxygen triple-isotope composition ($\Delta'^{17}O = \delta'^{17}O - \theta \times \delta'^{18}O$) of tropospheric O_2 is an important parameter used in mass-balance proxy estimates of gross oxygen productivity in the modern ocean and the global biosphere. We created a chemical reaction network box model of the $\Delta'^{17}O$ budget of tropospheric O_2 to examine the key controls on the oxygen triple-isotope composition of tropospheric O_2 ($\Delta'^{17}O$ $O_{2,\ trop}$). Our model is composed of four boxes: the stratosphere, the troposphere, the terrestrial biosphere/hydrosphere, and the marine biosphere/hydrosphere. We find that including an O_2 consumption flux via Mehler-like reactions in marine cyanobacteria equal to between 40 and 50% of marine gross oxygen productivity resolves three issues at once: 1) interlaboratory disagreements on the oxygen triple-isotope signature of tropospheric O_2 in the present day, 2) the incompatibility of model predictions of $\Delta'^{17}O$ $O_{2,\ trop}$ with corresponding air observations in two laboratory reference frames, and 3) puzzling discrepancies between concurrently measured $\Delta'^{17}O$-gross oxygen productivity and ^{14}C-net carbon productivity rates in the ocean. We also discuss how variations in the extent of Mehler-like reactions complicate $\Delta'^{17}O$-based interpretations of global gross oxygen productivity since the Last Glacial Maximum and propose a series of experiments to test our hypothesized large flux of O_2 being consumed by Mehler-like reactions in the global O_2 budget. Fundamentally, this work questions what variations in the $\Delta'^{17}O$ of O_2 indicate about global biogeochemical cycling.

1 Introduction

The oxygen triple isotope composition, $\Delta'^{17}O$, of O_2 quantifies the deviation of measured $\delta'^{17}O$ values from those predicted by mass-dependent fractionation. Mass-dependent reactions cause $\delta'^{17}O$ to change about ~0.5 times as much as $\delta'^{18}O$, while mass-independent reactions cause $\delta'^{17}O$ to vary with ratios much larger than 0.5 relative to $\delta'^{18}O$. The $\Delta'^{17}O$ of O_2 in the ocean and the atmosphere reflect the balance of stratospheric photochemistry and photosynthesis. $\Delta'^{17}O$ measurements of O_2 stored in these reservoirs can be used to calculate rates of gross oxygen productivity after being normalized with respect to the oxygen triple-isotope effects of biological O_2 uptake. This isotopic mass-balance approach has been used to estimate rates of marine gross oxygen productivity using O_2 dissolved in seawater as well as rates of global gross oxygen productivity on glacial-interglacial timescales using atmospheric O_2 trapped in glacial ice.[1,2]

The oxygen triple-isotope composition of tropospheric O_2 is a key end-member used in these calculations, so understanding why and how this parameter has the $\Delta'^{17}O$ composition that it does is critical for understanding the factors that affect biological productivity estimates using this proxy. Concerningly, laboratories working in the field disagree on the $\Delta'^{17}O$ of tropospheric O_2. Most laboratories measure $\Delta'^{17}O$ $O_{2,\,trop}$ to be around -0.441 ± 0.012‰, while the Hebrew University laboratory gauges it to be -0.507‰.[3–7] If the laboratories working in this field do not concur in their measurements of the isotopic composition of air, perhaps the most analyzed standard there is, what uncertainties does this introduce into applications of $\Delta'^{17}O$ as a proxy of gross oxygen productivity?

Work from our laboratory indicates that the parametrization for the oxygen triple-isotope effects of dark respiration measured at Hebrew University was compromised by analytical artifacts, and that a steeper mass-dependence for dark respiration better fits results of theoretical calculations and clumped oxygen isotope constraints.[8,9] If true, this low bias in the mass-dependent fractionation of dark respiration could have caused past marine gross oxygen productivity estimates to be overestimated by around 40%.[4,9] This observation also increases uncertainties in $\Delta'^{17}O$-based estimates of global gross oxygen productivity, because the oxygen triple-isotope effects of dark respiration is a large contributor to the overall oxygen triple-isotope effects of global biological O_2 uptake.

We created a chemical reaction network box model of the oxygen triple-isotope budget of tropospheric O_2 to explore these questions. Tropospheric O_2 has a negative $\Delta'^{17}O$ signal that mostly reflects the balance of stratospheric photochemistry and biological O_2 uptake. The creation of O_3 in the stratosphere imparts a positive $\Delta'^{17}O$ signal in O_3.[10] $O(^1D)$ produced by the photodissociation of O_3 is then quenched by stratospheric CO_2, which transfers the positive $\Delta'^{17}O$ signal in $O(^1D)$ to stratospheric CO_2. Stratospheric CO_2 is then transported into the troposphere by stratosphere-troposphere mixing where it exchanges oxygen isotopes with the hydrosphere. This transfers the positive $\Delta'^{17}O$ signal stored in tropospheric CO_2 to water, which leaves tropospheric O_2 depleted in $\Delta'^{17}O$. Oxygen isotope fractionation during biological O_2 consumption recycles some of the negative $\Delta'^{17}O$ derived from stratospheric chemistry and imparts its own $\Delta'^{17}O$ signature via slight variations in the mass-dependent fractionation of different biological O_2 uptake pathways.[11]

2 Background

2.1 Notation

Throughout this paper we use the linearized form of $\delta'^{18}O$ and $d'^{17}O$:

$$\delta'^{18}O = 1000 \ln \left(\frac{^{18}R_{sample}}{^{18}R_{VSMOW}} \right)$$

$$\delta'^{17}O = 1000 \ln \left(\frac{^{17}R_{sample}}{^{17}R_{VSMOW}} \right)$$

The molar ratios of ^{18}O and ^{17}O relative to ^{16}O are:

$$^{18}R = \frac{^{18}O}{^{16}O}$$

$$^{17}R = \frac{^{17}O}{^{16}O}$$

$^{18}R_{VSMOW}$ and $^{17}R_{VSMOW}$ are:

$$^{18}R_{VSMOW} = 0.0020052$$

$$^{17}R_{VSMOW} = 0.0003799$$

Fractionation factors $^{18}\alpha$ and $^{18}\alpha$ are defined as:

$$^{18}\alpha = \frac{^{18}R}{^{18}R_0}$$

$$^{17}\alpha = \frac{^{17}R}{^{17}R_0}$$

These fractionation factors are related by θ, a parameter that quantifies the oxygen-triple isotope effects of a given process, in the following way:

$$^{17}\alpha = \left(^{18}\alpha\right)^{\theta}$$

θ is calculated as follows:

$$\theta = \frac{\ln(^{17}\alpha)}{\ln(^{18}\alpha)}$$

Other fractionation factors used in this work are defined as follows:

$$^{18}\epsilon = (^{18}\alpha - 1) \times 1000$$

$$^{17}\epsilon = (^{17}\alpha - 1) \times 1000$$

$\Delta'^{17}O$ is defined as the following:

$$\Delta'^{17}O = \delta'^{17}O - \theta_{reference} \times \delta'^{18}O$$

We take $\theta_{reference}$ to be equal to be the slope characterizing terrestrial hydrology:[3]

$$\theta_{equil.} = 0.528$$

2.2 Biological O_2 uptake pathways

Oxygen consumption in the biosphere is dominated by dark respiration, but respiration is not the only O_2 consumption process relevant to the global O_2 budget. There are other oxygen consumption mechanisms that happen outside of mitochondria to manage reactive oxygen species or to limit the negative impacts of excess energy produced by the action of light on photosynthetic machinery which have not been fully accounted for in our understanding of the isotopic budget of tropospheric O_2.

2.2.1 Dark respiration

Dark respiration occurs through two processes: O_2 consumption by the cytochrome c oxidase pathway and O_2 consumption by the alternative oxidase pathway. Cytochrome c oxidase is the last enzyme in the respiratory electron transport chain, and is integral to the functioning of cellular metabolism in prokaryotes and eukaryotes.[12] It is understood to be the dominant biological O_2 uptake pathway globally. Blunier et al. (2002) used empirical measurements of carbon fluxes to estimate that the cytochrome C oxidase pathways accounts for 68% of the global O_2 consumption flux in the modern.[13] We note that this estimate refers specifically to the consumption of biomass, not to the consumption of oxygen, and is thus not necessarily representative of the amount of O_2 flowing through the cytochrome c oxidase pathway. The second process involved in dark respiration is the alternative oxidase pathway, which limits the production of reactive oxygen species by balancing rates of electron flow with rates of carbon fixation during photosynthesis.[14] The flux of O_2 through the alternative oxidase pathway is estimated to be roughly 10% of the flux moving through the cytochrome c oxidase pathway.[15]

2.2.2 Photorespiration

Photorespiration refers to the process where Rubisco catalyzes the addition of O_2 to RuBP instead of the addition of CO_2 to RuBP which is the desired reaction. Photorespiration creates intermediate metabolites that are toxic to the Calvin-Benson cycle which are then recycled into non-toxic forms to limit damage to cellular components.[16] The carboxylation of RuBP is a crucial step in photosynthesis, and the oxygenation of RuBP/the recycling of toxic intermediates wastes energy that would have been used to fix CO_2 into biomass. Up to 25% of the energy directed to fixing CO_2 is instead used to add O_2 to RuBP which dramatically reduces the efficiency of photosynthesis.[17] Photorespiration it is the second most important O_2 consumption process in the global biosphere after the cytochrome c oxidase pathway.[18] Blunier et al (2002) used empirical measurements of carbon fluxes to estimate that photorespiration accounts for 24% of the global O_2 consumption flux in the modern.[13] Rates of photorespiration are high in cellular environments with high O_2 concentrations/low CO_2 concentrations. These conditions are abundant in C_3 plants, but are rarely encountered in C_4 plants or in marine photosynthesizers. C_4 plants and marine photosynthesizers have carbon concentrating mechanisms that increase the concentration of CO_2 around the active site of RuBP which limit the potential for O_2 react with RuBP.[19] Photorespiration helps modulate electron flow to prevent photoinhibition in plants experiencing drought, salinity, and light stress, all of which expose cells to deleterious photon fluxes relative to potential rates of carbon fixation.[20]

2.2.3 O_2 photoreduction processes

Flavodiiron proteins are widespread modular enzymes in bacteria, archaea, and eukarya that detoxify O_2 and NO in cells.[21] In higher plants, cyanobacteria, and algae these proteins are crucial catalysts of reactions that protect cells from excessive and variable light levels.[22] We focus on the two most important O_2 reduction pathways: the Mehler reaction and Mehler-like reactions. The Mehler reaction refers to the reduction of O_2 produced at photosystem I to superoxide, hydrogen peroxide, and then eventually water, which dissipates excess energy from sunlight and keeps cells from getting “sunburned”.[23] This process is also referred to as the “water-water” cycle, because electrons from water in photosystem II are donated to O_2 for the reduction of O_2 at photosystem I, which results in no net flux of O_2.[24] The biochemical mechanism by which the Mehler reaction occurs is not fully understood. Badger et al. (2000), a widely-cited literature review of experimental work quantifying rates of O_2 photoreduction in photosynthetic organisms, indicated that the Mehler reaction accounts for ~10% of the global biological O_2 consumption flux.[13,25] The distinguishing feature between the Mehler reaction and Mehler-like reactions is that the former produces reactive oxygen species as intermediate species in the conversion of O_2 to H_2O, while Mehler-like reactions directly reduce O_2 to H_2O.[26] Another important difference between the two is that the Mehler reaction is limited to higher plants, including both C_3 and C_4 photosynthesizers, while Mehler-like reactions are found in marine photosynthetic organisms.[21] We will refer to these reactions as the 'terrestrial Mehler reaction' and 'marine Mehler-like reactions' throughout the rest of this work.

Previous work examining the oxygen triple-isotope budget of tropospheric O_2 accounted for the isotopic effects of the terrestrial Mehler reaction, but neglected the isotopic effects of marine Mehler-like reactions.[13] This is strange, given that extant literature indicates that marine Mehler-like reactions likely constitute a large O_2 consumption flux in the ocean. Goosney and Miller (1997) observed that O_2 chlorophyll quenching was between 50 and 70% of the corresponding CO_2 flux in photoautotrophically grown *Synechocystis* which indicates that high rates of marine Mehler-like reactions occur in this model organism of marine cyanobacteria.[27] Badger et al. (2000) found that marine Mehler-like reactions could consume up to ~50% of photosynthetic electrons in marine cyanobacteria.[25] Isotopic evidence from Helman et al. (2005) indicated that up to 40% of electron flow out of photosystem I in *Synechocystis* could be consumed by marine Mehler-like reactions.[28] Roberty et al. (2014) found that up to 50% of electron flow in *Symbiodinium*, a dinoflagellate associated with coral reefs, could be consumed by marine Mehler-like reactions.[29] Allahverdiyeva et al. (2011) observed that 20% of electrons flowing out of photosystem I in *Synechocystis* are consumed by marine Mehler-like reactions ina bicarbonate rich environment, whereas under conditions of bicarbonate deprivation up to 50-60% of gross photosynthetic O_2 could be consumed by marine Mehler-like reactions.[30] Schuurmans et al. (2015) found that marine Mehler-like reactions consume around 34% of the total O_2 evolved at photosystem II in *Synechocystis*. *In-vivo* experiments from Theyne et al. (2021) indicate that marine Mehler-like reactions consume about 35% of photosynthetic electron flow in *Synechocystis*.[31] The large rates of O_2 photoreduction in marine cyanobacteria observed in these studies suggest that marine Mehler-like reactions play an important role in marine biological O_2 uptake which needs be accounted for in the $\Delta'^{17}O$ budget of tropospheric O_2.

2.2 Outstanding issues in the $\Delta'^{17}O$ literature

In this paper we address three unresolved issues in the interpretation of $\Delta'^{17}O$-gross oxygen productivity rates: 1) interlaboratory disagreements on the $\Delta'^{17}O$ signature of tropospheric O_2, 2) uncertainties induced by recent work showing that $\theta_{\text{dark resp.}}$ is more variable than previously believed, and 3) curious discrepancies between concurrent measurements of $\Delta'^{17}O$-gross oxygen productivity and ^{14}C-net carbon productivity rates in the ocean.

2.2.1 Interlaboratory disagreement about the $\Delta'^{17}O$ of tropospheric O_2

Figure 1 shows the $\delta'^{18}O$ and $\Delta'^{17}O$ values of tropospheric O_2 measured by Luz and Barkan (2011) at the Hebrew University of Jerusalem, by Young et al. (2014) at the University of California, Los Angeles, UCLA, by Yeung et al. (2018) at Rice University, by Pack et al. (2017) at the University of Göttingen, and by Wostbrock et al. (2020) at the University of New Mexico, UNM.[32,3,33,5,6] Observations from Rice University and UCLA were calibrated to the isotopic composition of San Carlos Olivine measured at UNM such that they are in the same reference frame as the other data shown. This conversion decreased their respective $\Delta'^{17}O$ values by ~0.2‰. The data from UCLA, Rice University, and the University of Göttingen are all mostly within the uncertainties of the $\Delta'^{17}O$ $O_{2,\text{ trop}}$ observation from UNM, which we believe to be the most up-to-date and methodologically sound measurement available in the literature. We consider the $\Delta'^{17}O$ $O_{2,\text{ trop}}$ observation from UNM to be representative of the measurements from UCLA, Rice University, and the University of Göttingen. The $\Delta'^{17}O$ $O_{2,\text{ trop}}$ observation from Hebrew University is lower than all other measurements by ~0.05‰, which is quite a large offset in oxygen-triple isotope space. The probable reason for this discrepancy between the measurements is that Ar was stripped from O_2 samples prior to isotopic analysis conducted at UCLA, Rice University, the University of Göttingen, and UNM while O_2 and Ar were run together during isotopic analysis at Hebrew University.

The fact O_2 was run with Ar during the measurement of the divergent $\Delta'^{17}O$ observation from Hebrew University is important because pressure-baseline effects caused by excess gas entering the mass spectrometer source, in the form of Ar in this case, can cause observed $\Delta'^{17}O$ values to be inaccurate on the order of ~0.04‰.[4] This error is comparable in magnitude to the difference between measurements of $\Delta'^{17}O$ $O_{2,\,trop}$ that ran O_2 without Ar and the measurement that ran O_2 with Ar. The interlaboratory disagreement on the oxygen triple-isotope signature of tropospheric O_2 is disturbing because it is one of three key parameters necessary to calculate $\Delta'^{17}O$-based rates of gross oxygen productivity alongside the oxygen triple-isotope composition of photosynthetically-derived O_2 and the $\theta_{O2\ uptake}$ value characterizing dark respiration.[1] Most of the parameters used to calculate $\Delta'^{17}O$-gross oxygen productivity rates in the literature come from the Hebrew University group, so the fact that the air observation from this laboratory is so different from the others suggests that analytical errors could have compromised other parameters necessary to the correct application of the $\Delta'^{17}O$ proxy other than the isotopic composition of tropospheric O_2.

2.2.2 $\theta_{dark\ resp.}$ is less certain than previously believed

The $\theta_{O2\ uptake}$ value characterizing biological O_2 uptake in the surface ocean is also an important parameter crucial in applying the $\Delta'^{17}O$-GOP proxy correctly. Choosing the appropriate value for $\theta_{dark\ resp.}$ leads the $\Delta'^{17}O$ of observed O_2 samples to be insensitive to the isotopic effects of biological O_2 uptake, and thus to simply reflect a mass-balance between tropospheric O_2 and photosynthetic O_2. If this value is parameterized incorrectly, then productivity estimates made using the oxygen triple-isotope system would reflect the isotopic effects of O_2 consumption processes as well as O_2 production processes. In prior applications of the $\Delta'^{17}O$ proxy for marine gross oxygen productivity, dark respiration was assumed to be the only O_2 consumption process relevant in the surface ocean. This means that the overall oxygen triple-isotope effects of O_2 consumption in the photic zone was set to be equal to the $\theta_{O2\ uptake}$ value of dark respiration observed at Hebrew University which lies between 0.514 to 0.516.[8,15] $\theta_{dark\ resp.}$ was found to be invariant among microbes it was measured in despite large differences in $^{18}\alpha$ fractionation factors among the different species.[34] This is a puzzling because the species-dependent expression of O_2 consumption pathways in dark respiration that cause variability in $^{18}\alpha$ values should, in principle, modify $\theta_{dark\ resp.}$ values as well. To estimate rates of global gross oxygen productivity, $\theta_{O2\ uptake}$ is not simply set to $\theta_{dark\ resp.}$ because of the importance of other O_2 consumption processes such as photorespiration, the alternative oxidase pathway, O_2 photoreduction processes etc. in the terrestrial and marine biospheres. Nonetheless, the oxygen triple-isotope effects of dark respiration represent a large proportion of global $\theta_{O2\ uptake}$ due to the dominance of dark respiration at the global scale, meaning inaccuracies in $\theta_{dark\ resp.}$ impact global gross oxygen productivity estimates also.

Recent work from our research group indicates that the $\theta_{O2\ uptake}$ value of dark respiration used in past $\Delta'^{17}O$-gross oxygen productivity estimates, $\theta_{COX} = 0.516 \pm 0.001$, was measured inaccurately due to analytical artifacts that cause it to be lower than the true value. Ash et al. (2020) measured $\theta_{dark\ resp.}$ to be 0.520 ± 0.001 in microbial communities sourced from Lake Houston, which is larger than the earlier observation by 0.004.[9,15] Theoretical calculations of the oxygen triple-isotope and clumped oxygen isotope effects of the proteins involved in respiratory oxygen isotope fractionation predict a steeper mass-dependence for $\theta_{dark\ resp.}$ which is in-line with the new observation. Additionally, Figure 2 shows upcoming model results from our research group which indicate that the higher $\theta_{dark\ resp.}$ better explains $\Delta'^{17}O$ measurements of dissolved O_2 in the deep ocean than the lower $\theta_{dark\ resp.}$ does.

The fundamental assumption that $\theta_{dark\ resp.}$ is invariant has also been questioned by recent work. Stolper et al. (2018) found that $\theta_{dark\ resp.}$ increased from 0.508 to 0.513 as temperature rose from 15 °C to 37 °C in experiments conducted on *Escherichia coli*, and hypothesized that this relationship was due to a shifting balance between O_2 enzyme-binding and O_2 reduction pathways.[35] Figure 3 shows the temperature dependence of $\theta_{dark\ resp}$ observed in the study. While the $\theta_{dark\ resp.}$ values observed in this work are much lower than what was measured in Ash et al. (2020), the key point to get from this research is that $\theta_{dark\ resp.}$ could vary depending on the environmental conditions under which O_2 consumption processes occur.

These findings indicate that the proper value for $\theta_{O2\ uptake}$ to use when applying the $\Delta'^{17}O$-gross oxygen productivity proxy is less simple than previously believed. While the variations in $\theta_{dark\ resp.}$ we have discussed may seem negligible, they are actually quite significant. Small variations in the $\theta_{dark\ resp.}$ value used during mass-balance calculations cause dramatic changes in predicted gross oxygen productivity rates. For every 0.001 error in $\theta_{dark\ resp.}$, calculated marine gross oxygen productivity increase in error by 10-15%. Scaling this relationship to the observed 0.004 error in the value of $\theta_{dark\ resp.}$ indicates that previous estimates of marine gross oxygen productivity could be around ~40-60% too high.[9] The temperature dependence of $\theta_{dark\ resp.}$ observed in Stolper et al. (2018)suggests that the assumption that $\theta_{dark\ resp.}$ is static may not be appropriate. This means that environmentally forced changes in $\theta_{O2\ uptake}$ values could contribute to variations in observed $\Delta'^{17}O$ values which would limit interpretations of changes in $\Delta'^{17}O$ as being due solely to increases or decreases in gross oxygen productivity.

2.2.3 Observed Δ'^{17}O-GOP rates do not make sense physiologically

An unanswered conundrum in the Δ'^{17}O-gross oxygen productivity literature is the origin of large discrepancies between concurrently measured Δ'^{17}O-based estimates of gross oxygen productivity and ^{14}C-based estimates of net carbon productivity at sites from around the global ocean. The Δ'^{17}O-gross oxygen productivity method is an incubation-independent tracer that captures *in-situ* rates of gross oxygen productivity integrated over the residence time of O_2 in the mixed layer, which is typically on the order of ~1-2 weeks, while the ^{14}C-incorporation method is an incubation-dependent proxy that captures *in-vitro* rates of net carbon productivity during the length of the incubation, which is typically anywhere between 6 and 24 hours long.[1,36] The Δ'^{17}O proxy tracks the total amount of oxygen produced during photosynthesis, i.e. does not account for O_2 consumed by photosynthesizers, while the ^{14}C method tracks the amount of fixed biomass that leaves photosynthetic microbes and enters the environment, i.e. accounts for carbon biomass consumed photosynthesizers. Figure 4 is a plot of concurrently measured Δ'^{17}O-gross oxygen productivity data versus ^{14}C-net carbon productivity data as shown in Juranek et al. (2013).[1] A line with slope equal to 3.3 is plotted alongside this data, which represents the physiological upper limit of the ratio of gross oxygen productivity to net carbon productivity for aquatic photosynthesizers as measured in Halsey et al. (2010).[37]

Five of the six ratios of gross oxygen productivity to net carbon productivity observed in the global ocean lie above the physiological limit discussed above. How is this possible? Does the large variability in these ratios represent fundamental differences between oxygen and carbon cycling in the surface ocean, over and beyond what is observed in the laboratory? We know that oxygen production and carbon fixation are linked by photosynthesis, but they are not necessarily proportional to one another. This is due to the activity of O_2 consumption processes other than mitochondrial respiration, like photorespiration or O_2 photoreduction mechanisms, which are not directly involved in photosynthesis but alter the flux of O_2 in the cell without contributing to C cycling. The relative expression of these O_2 consumption processes likely vary on different spatial and temporal scales depending on environmental conditions, which could lead to local differences in the relative rates of oxygen and carbon cycling. Another potential contributor to the large variability in these ratios are methodological biases in either the $\Delta'^{17}O$-gross oxygen productivity or ^{14}C-net carbon productivity proxies, or both. While the $\Delta'^{17}O$-gross oxygen productivity proxy quantifies photosynthesis rates on weekly timescales and integrates gross oxygen productivity occurring at the field scale, the ^{14}C-net carbon productivity proxy quantifies photosynthesis rates on hourly timescales and integrates net carbon productivity occurring in the incubation bottle. These differences lead the proxies to track O_2 and C production on qualitatively distinct spatial and temporal scales.

3 Methods

3.1 Model description

We created a box model of the oxygen-triple isotope budget of oxygen in the troposphere to explore the questions raised above. It is a modified version of the chemical reaction network model used in Young et al. (2014).[3] Our model is composed of four boxes: the stratosphere, the troposphere, the terrestrial biosphere/hydrosphere, and the marine biosphere/hydrosphere. The terrestrial and marine hydrospheres are considered to be effectively infinite reservoirs of O_2 that are characterized by distinct source water oxygen isotope compositions. Figure 5 is a schematic diagram of the model. Reactions necessary to model the oxygen chemistry of the atmosphere were formulated as differential equations which were solved in Python. Figures were also produced in Python. We used the pandas, numpy, scipy, and matplotlib packages to conduct this work. The code for the base model and the different model runs will be made available on GitHub when the paper resulting from this work is published. Outputs of the model include the isotopic composition and fluxes of O_2, O, O(^{1}D), O_3 and CO_2 in and among the reservoirs. Table 1 lists the chemical species tracked by the model.

3.2 Stratospheric chemistry

Stratospheric photochemistry is an important driver of Δ'^{17}O $O_{2,\ trop}$. The creation of O_3 in the stratosphere generates a positive Δ'^{17}O signal in O_3, as it fractionates oxygen isotopes mass-independently.[10] O(^{1}D) is produced by the photodissociation of O_3 which in turn is quenched by stratospheric CO_2, transferring the positive Δ'^{17}O signal from O(^{1}D) originating in stratospheric O_3 to stratospheric CO_2. Stratospheric CO_2 is then transported into the troposphere through stratosphere-troposphere mixing where it exchanges oxygen isotopes with the hydrosphere, depositing the positive Δ'^{17}O signal in water which contributes to the depleted Δ'^{17}O signal in tropospheric O_2.

Most of the rate constants used in the model were calculated using expressions and parameter values from Burkholder et al. (2020).[38] Where temperature, actinic flux, and number-density dependencies are needed for rate constant calculations we use values appropriate for conditions at 25 km. We do this because ozone production and concentration peak at this altitude. The temperature at 25 km this height is 220 K, while the number density of the stratosphere is 8.3×10^{17} cm^{-3}. Table 2 lists all the reactions included in the model alongside their rate constants, we refer to isotopologue specific rate constants using the naming scheme contained in this table.

3.2.1 O_2 Photolysis

Stratospheric oxygen chemistry begins with the photolysis of O_2 which produces atomic oxygen:

$$\text{(R1A) } O_2 + h\nu \rightarrow O + O$$

$$\text{(R1B) } O^{18}O + h\nu \rightarrow O + {}^{18}O$$

$$\text{(R1C) } O^{17}O + h\nu \rightarrow O + {}^{17}O$$

The rate constants for reactions R1A-R1C were determined using the standard photolytic rate constant expression:

$$J = \int_{\lambda} \sigma(\lambda)\phi(\lambda)f(\lambda)d\lambda$$

Where $\sigma(\lambda)$ is the absorption cross-section (cm^2), $\varphi(\lambda)$ is the quantum yield of the reaction, and $f(\lambda)$ is the actinic flux ($cm^{-2}\ s^{-1}$). Absorption cross-section data were obtained from Sander et al. (2006) and we used a typical actinic flux for ~25 km at a 30° solar zenith angle reported in DeMore et al. (1997).[39,40]

3.2.2 O_3 Creation

O_2 interacts with O and a collision partner to form O_3. These reactions are the only source of oxygen-triple isotope effects in the stratosphere:

$$\text{(R2A) } O_2 + O + M \rightarrow O_3$$

$$\text{(R2B) } O_2 + {}^{18}O + M \rightarrow OO^{18}O$$

$$\text{(R2C) } O_2 + {}^{17}O + M \rightarrow OO^{17}O$$

$$\text{(R2D) } O^{18}O + O + M \rightarrow OO^{18}O$$

$$\text{(R2E) } O^{17}O + O + M \rightarrow OO^{17}O$$

In these reactions M represents unspecified collision partners (either N_2 or O_2). We calculate the rate constant for R2A using the low-pressure limit expression for termolecular reactions discussed in Burkholder et al. (2020) which is appropriate for conditions at 25 km:[38]

$$k_o(T) = k_o^{298}\left(\frac{T}{298K}\right)^{-m}[M]$$

Where k_o^{298} is the rate constant at 298 K, n is a fitting parameter, and [M] is the number density of the atmosphere for a given altitude. We use 8.3×10^{17} cm^{-3}, the number density of the stratosphere at 25 km, for [M]. R2B-R2E are multiplied by α_{MIF}, which quantifies the degree of mass independent fractionation during ozone production. We tune this value to match the targeted $\Delta'^{17}O$ of stratospheric O_3 and find that a value of 1.065 suits this purpose well.

3.2.3 O_3 Photolysis

O_3 photolysis produces O with high $\Delta'^{17}O$ values:

(R3A) $O_3 + h\nu \rightarrow O_2 + O$

(R3B) $OO^{18}O + h\nu \rightarrow O_2 + {}^{18}O$

(R3C) $OO^{18}O + h\nu \rightarrow O^{18}O + O$

(R3D) $OO^{17}O + h\nu \rightarrow O_2 + {}^{17}O$

(R3E) $OO^{17}O + h\nu \rightarrow O^{17}O + O$

O_3 photolysis also produces $O(^1D)$ with high $\Delta'^{17}O$ values:

(R4A) $O_3 + h\nu \rightarrow O_2 + O(^1D)$

(R4B) $OO^{18}O + h\nu \rightarrow O_2 + {}^{18}O(^1D)$

(R4C) $OO^{18}O + h\nu \rightarrow O^{18}O + O(^1D)$

(R4D) $OO^{17}O + h\nu \rightarrow O_2 + {}^{17}O(^1D)$

(R4E) $OO^{17}O + h\nu \rightarrow O^{17}O + O(^1D)$

Quantum yields for O vs. $O(^1D)$ are taken from Matsumi and Kawasaki (2003) where:[41]

$$\phi_{O(^1D)}(\lambda) + \phi_O(\lambda) = 1$$

and $\varphi_{O(1D)}(\lambda)$ has a value of 0.9 for $\lambda < 306$ nm and 0 for $\lambda > 340$ nm. We evaluated equations R3A-R4E using absorption cross section data from Sander et al. (2006) and the same actinic flux for ~25 km at a solar zenith angle of 30° from DeMore et al. (1997) used to calculate R1A-C.[39,40]

O_3 reacts with O to produce O_2:

(R5A) $O_3 + O \rightarrow O_2 + O_2$

(R5B) $OO^{18}O + O \rightarrow O_2 + O^{18}O$

(R5C) $OO^{17}O + O \rightarrow O_2 + O^{17}O$

(R5D) $O_3 + {}^{18}O \rightarrow O_2 + O^{18}O$

(R5E) $O_3 + {}^{17}O \rightarrow O_2 + O^{17}O$

We use the modified Arrhenius equation recommended in Burkholder et al. (2020) to determine rate constants for these reactions:[38]

$$k(T) = A\left(\frac{T}{298K}^{n}\right)\left(e^{\frac{-E}{RT}}\right)$$

Where A is the Arrhenius A-factor, T is the temperature in Kelvin, n is a fitting parameter and E/R is the recommended temperature dependence or "activation temperature". R5A is multiplied by ½ to calculate R5B-E which accounts for the probability that ^{18}O or ^{17}O is sequestered in either of the two O_2 molecules produced by the reaction.

O_3 also reacts with $O(^1D)$ to produce O_2:

(R6A) $O_3 + O(^1D) \rightarrow O_2 + O_2$

(R6B) $OO^{18}O + O(^1D) \rightarrow O_2 + O^{18}O$

(R6C) $OO^{17}O + O(^1D) \rightarrow O_2 + O^{17}O$

(R6D) $O_3 + {}^{18}O(^1D) \rightarrow O_2 + O^{18}O$

$$(R6E)\ O_3 + {}^{17}O({}^1D) \rightarrow O_2 + O^{17}O$$

$$(R6F)\ O_3 + O({}^1D) \rightarrow O_2 + O + O$$

$$(R6G)\ OO^{18}O + O({}^1D) \rightarrow O_2 + O + {}^{18}O$$

$$(R6H)\ OO^{18}O + O({}^1D) \rightarrow O^{18}O + O + O$$

$$(R6I)\ OO^{17}O + O({}^1D) \rightarrow O_2 + O + {}^{17}O$$

$$(R6J)\ OO^{17}O + O({}^1D) \rightarrow O^{17}O + O + O$$

$$(R6K)\ O_3 + {}^{18}O({}^1D) \rightarrow O_2 + O + {}^{18}O$$

$$(R6L)\ O_3 + {}^{18}O({}^1D) \rightarrow O^{18}O + O + O$$

$$(R6M)\ O_3 + {}^{17}O({}^1D) \rightarrow O_2 + O + {}^{17}O$$

$$(R6N)\ O_3 + {}^{17}O({}^1D) \rightarrow O^{17}O + O + O$$

R6A was calculated using the same procedure as R5A but with parameters appropriate for this reaction. R6A is multiplied by ½ to determine R6B-R6E which accounts for the fact that ${}^{17}O$ or ${}^{18}O$ can be sequestered in either of the two O_2 molecules produced by the reaction. R6A is multiplied by ½ × ½ to determine R6F-R6J to account for the fact that ${}^{17}O$ or ${}^{18}O$ can be sequestered in either O_2 or one of the two O species produced.

3.2.4 O(¹D) Quenching by O_2

$O({}^1D)$ reacts with a collision partner to form O_2 and O:

$$(R7A)\ O_2 + O({}^1D) \rightarrow O_2 + O$$

$$(R7B)\ O_2 + {}^{18}O({}^1D) \rightarrow O_2 + {}^{18}O$$

$$(R7C)\ O_2 + {}^{18}O({}^1D) \rightarrow O^{18}O + O$$

$$(R7D)\ O_2 + {}^{17}O({}^1D) \rightarrow O_2 + {}^{17}O$$

$$(R7E)\ O_2 + {}^{17}O({}^1D) \rightarrow O^{17}O + O$$

The base reaction rate for R7A was calculated using the same procedure as R6A, but with parameters appropriate for this reaction. To account for the fact that N_2 is also a collision partner which $O(^1D)$ can react with, we apply a correction factor that uses a steady-state assumption for the concentrations of O_2 and N_2 in the atmosphere (21% and 78% respectively). We multiply the base reaction rates by a correction factor which takes the following form:

$$\frac{0.78 + 0.21}{0.21}$$

3.2.5 O_2 Isotope Exchange

O_2 exchanges oxygen isotopes with $O(^1D)$:

$$\text{(R8A) } O_2 + {}^{18}O \rightarrow O^{18}O + O$$

$$\text{(R8B) } O^{18}O + O \rightarrow O_2 + {}^{18}O$$

$$\text{(R8C) } O_2 + {}^{17}O \rightarrow O^{17}O + O$$

$$\text{(R8D) } O^{17}O + O \rightarrow O_2 + {}^{17}O$$

We use the experimentally-constrained expression of Fleurat-Lessard et al. (2003) to calculate the rate constant for this reaction:[42]

$$k = (2.7 \pm 0.4) \left(\frac{300K}{T} \right)^{0.9 \pm 0.5}$$

Where T is temperature in Kelvin. R8B and R8D are multiplied by ½ to account for the fact that the ^{17}O or ^{18}O is transferred from the O_2 isotopologue to O.

3.2.6 $O(^1D)$ Quenching by CO_2

$O(^1D)$ is quenched by CO_2 which transfers the positive $\Delta'^{17}O$ signal created during O_3 formation to stratospheric CO_2:

$$\text{(R9A) } CO_2 + O(^1D) \rightarrow CO_2 + O$$

$$\text{(R9B) } CO_2 + {}^{18}O(^1D) \rightarrow CO_2 + {}^{18}O$$

$$\text{(R9C) } CO_2 + {}^{18}O({}^{1}D) \rightarrow CO_2 + {}^{18}O$$

$$\text{(R9D) } CO^{18}O + O({}^{1}D) \rightarrow CO_2 + {}^{18}O$$

$$\text{(R9E) } CO^{18}O + O({}^{1}D) \rightarrow CO^{18}O + O$$

$$\text{(R9F) } CO_2 + {}^{17}O({}^{1}D) \rightarrow CO_2 + {}^{17}O$$

$$\text{(R9G) } CO_2 + {}^{17}O({}^{1}D) \rightarrow CO^{17}O + O$$

$$\text{(R9H) } CO^{17}O + O({}^{1}D) \rightarrow CO_2 + {}^{17}O$$

$$\text{(R9I) } CO^{17}O + O({}^{1}D) \rightarrow CO^{17}O + O$$

Young et al. (2014) cited Yung et al. (1991) to obtain an initial value for the rate constant of R9A of 4.6×10^{-11} cm^3 $second^{-1}$ with the base rate constants for R9B-R9I being multiplied by ½ to account for the probability of ^{17}O or ^{18}O being sequestered in CO_2 or $O(^1D)$.[3,10] This rate constant value and fraction actually referred to the quenching of $O(^1D)$ by CO:

$$CO + O({}^{1}D) \rightarrow CO + O$$

The quenching of $O(^1D)$ by CO_2 includes a CO_3 intermediate such that:

$$O({}^{1}D) + CO_2 \rightarrow CO_3{}^{*}$$

$$CO_3{}^{*} \rightarrow CO_2 + O$$

Which means that instead of the ½ partitioning used in Young et al. (2014), the partitioning should be either ⅓ or ⅔ depending on whether ^{17}O or ^{18}O is sequestered in O or CO_2. The correct rate constant was calculated in the same way as R7A using parameters appropriate for this reaction.

3.3 CO_2-H_2O Isotope Exchange

In the original model, CO_2-H_2O isotope exchange occurred between the troposphere and a global biosphere/hydrosphere.[3] We split this global biosphere/hydrosphere box into terrestrial and marine components to account for their distinct isotopic effects. When stratospheric CO_2 is transported into the troposphere, it exchanges oxygen isotopes with the terrestrial hydrosphere:

(R10At) $CO_2 + H_2{}^{18}O \rightarrow CO^{18}O + H_2O$

(R10Bt) $CO^{18}O + H_2{}^{18}O \rightarrow CO_2 + H_2{}^{18}O$

(R10Ct) $CO_2 + H_2{}^{17}O \rightarrow CO^{17}O + H_2O$

(R10Dt) $CO^{17}O + H_2{}^{17}O \rightarrow CO_2 + H_2{}^{17}O$

and with the marine hydrosphere:

(R10Am) $CO_2 + H_2{}^{18}O \rightarrow CO^{18}O + H_2O$

(R10Bm) $CO^{18}O + H_2{}^{18}O \rightarrow CO_2 + H_2{}^{18}O$

(R10Cm) $CO_2 + H_2{}^{17}O \rightarrow CO^{17}O + H_2O$

(R10Dm) $CO^{17}O + H_2{}^{17}O \rightarrow CO_2 + H_2{}^{17}O$

Differences in the isotopic effects of the marine and terrestrial biosphere result from the fractionation of terrestrial waters by evapotranspiration, the isotopic enrichment in photosynthetic O_2 produced by marine phytoplankton, and equilibrium isotope fractionation between CO_2 and H_2O. Equilibrium oxygen isotope fractionation between CO_2 and H_2O is a function of temperature as shown in the following experimentally-determined expression modified from Beck et al. (2005):[43]

$$\alpha_{CO_{2(aq)}-H_2O} = e^{\frac{2.5\pm0.03(10^6T^{-2})+12.12\pm0.33}{1000}}$$

Which yields an $\alpha_{CO2\text{-}H2O}$ of 1.041 when evaluated at 25 °C.

Oxygen in CO_2 is estimated to have a turnover time of around 1 year with respect to the global biosphere.[44] R10Bt, R10Dt, R10Bm and R10Dm are thus equal to 1 year^{-1}.

The reaction rate of R10At was calculated as follows:

$$k_{R10At} = k_{R10Bt}\alpha_{CO_2-H_2O}(1.0045^{18}R_{VSMOW})$$

The reaction rate of R10Ct was calculated as follows:

$$k_{R10Ct} = k_{R10Bt}(\alpha_{CO_2-H_2O})^{\theta_{equil.}}(1.0045^{\theta_{transp.}} {}^{17}R_{VSMOW})$$

The isotopic effects of terrestrial evapotranspiration is an important model parameter as it is a key control on the $\delta'^{18}O$ of tropospheric O_2 along with kinetic isotope fractionation attending biological O_2 uptake. A widely used GIS-based estimate for the global average leaf water $\delta'^{18}O$ at sites of evaporation within leaves is 6.5‰ as calculated in West et al. (2008).[45] Farquhar et al. (1993) predicted an enrichment of 4.4‰ for this same parameter using a simple model of CO_2 isotope dynamics in leaves, which indicates that this value has an uncertainty on the order of at least 2.1‰.[46] We assign the enrichment in terrestrial source waters to be 4.4‰, the modeled value from Farquhar et al. (1993), which brings the $\delta'^{18}O$ of tropospheric O_2 in-line with our target value. We account for this fractionation by including 1.0045 in the equations quantifying CO_2-H_2O isotope exchange between the troposphere and the terrestrial biosphere. We calculate the oxygen triple-isotope effects of transpiration, $\theta_{transp.}$,using the following expression from from Landais et al. (2006):[47]

$$\theta_{transp.} = (-0.0078 \pm 0.0026) \times h + 0.5216 \pm 0.0008$$

We evaluate this equation for a global average relative humidity of ~80%, which is the value observed in the most productive terrestrial ecosystems and at the ocean's surface.[48] This gives us a $\theta_{transp.}$ value of 0.5154.

The reaction rate for R10Am was calculated as follows:

$$k_{R10Am} = k_{R10Bm}(\alpha_{CO_2-H_2O})^{18}R_{VSMOW}$$

The reaction rate for R10Cm was calculated as follows:

$$k_{R10Cm} = k_{R10Bm}(\alpha_{CO_2-H_2O})^{\theta_{equil.}} {}^{17}R_{VSMOW}$$

We scaled these rate constants by a transportation normalization factor to account for two factors missing from our model. The first is that the mass-independent fractionation which occurs during O_3 formation is temperature dependent, whereas in our approach we model stratospheric chemistry as occurring solely at a temperature of 220 K. The second is that O_3 photolysis appears to display mass-dependent isotope fractionation depending on the wavelength of light it interacts with, which is not included in this model formulation.[49] The combination of these two processes lead to latitudinal, altitudinal, and seasonal variations in the isotopic composition of stratospheric CO_2, with stratospheric transport and mixing redistributing air with lower $\Delta'^{17}O$ values for CO_2 produced in the tropics to the subtropics and onwards into the the polar vortex, which eventually homogenizes to an extratropical average.[49] In our model, on the other hand, the $\Delta'^{17}O$ of stratospheric CO_2 is assumed to be homogenous throughout the stratosphere. Our chemical reaction network reproduces steady-state isotopic values comparable to the stratosphere-only kinetics modeling in Table 2 of Wiegel et al. (2013), but does not include the impact of within-stratosphere variations which is why we apply the transportation normalization factor.[49] We tune the value of the transportation normalization factor so that the $\Delta'^{17}O$ of CO_2 in the stratosphere is as close to our target value as possible and find that a value of 0.027 works well for this purpose.

The rate constants calculated above were converted into the appropriate units as follows: Photolytic rate constants were converted to units of year^{-1} from units of second^{-1} by multiplying by the number of seconds in a year, 31,536,000 second year^{-1}. Kinetic rate constants were converted to units of moles year^{-1} from units of cm^3 second^{-1} by multiplying by the number of seconds in a year and Avogadro's number divided by the volume of the stratosphere, 2.8×10^{25} cm^3.

3.4 Stratosphere-troposphere mixing

Stratosphere-troposphere mixing determines the degree to which the positive $\Delta'^{17}O$ signal produced by stratospheric photochemistry is expressed in the troposphere. We use a stratosphere-troposphere transport rate of 6.8×10^{17} kg year^{-1}, equivalent to 2.4×10^{19} moles year^{-1}, which was determined in Appenzeller and Holton (1996).[50] The mass of the atmosphere is 5.2×10^{18} kg.[51] The stratosphere contains about ~10% of the total mass of the atmosphere, meaning the mass of the stratosphere is 5.2×10^{17} kg, equivalent to 1.8×10^{19} moles of air.[52] Dividing this value by the stratosphere-troposphere mixing rate yields a rate constant for stratosphere-troposphere mixing, $k_{S\text{-}T}$, of ~1 year^{-1}. Since the troposphere contains ~10x more mass than the stratosphere, continuity demands that the troposphere-stratosphere rate constant, $k_{T\text{-}S}$, be 0.1 year^{-1}.

3.4 Photosynthesis and respiration

We use a turnover time of tropospheric O_2 with respect to photosynthesis and respiration of 1.2 kyr to calculate the global respiration rate constant, k_r, of ~8×10^{-4} year^{-1}.[53] The global rate of photosynthesis, k_p, is tuned to reproduce the mixing ratio of atmospheric O_2, 21%, which gives us a value of 1.65×10^{-3} year^{-1} for this parameter. The ratio of photosynthesis to respiration, k_p/k_r, is defined as global gross oxygen productivity in this model formulation. Global gross oxygen productivity is held constant in all of the model runs discussed in this work.

Luz et al. (2014) made estimates for the overall $^{18}\varepsilon$ due to biological oxygen uptake in the marine and terrestrial biospheres.[54] We calculate the isotopic effects of marine biological O_2 uptake by taking a weighted average of estimates in their Table 6 for isotopic fractionation in the surface ocean and the deep ocean such that:

$$^{18}\epsilon_{marine} = \frac{(25.2 \pm 1.9) \times 0.33 + (23.2 \pm 0.5) \times 0.04}{0.37} = 25$$

The isotopic effects of respiratory fractionation are amplified by the O_2 enriched in $\delta'^{18}O$ produced by marine phytoplankton.[55] We determined the average $^{18}\varepsilon$ of marine photosynthesis to be 3.389‰ using data from Table 1 of Luz and Barkan (2011).[32] The combined isotopic effects of photosynthesis and respiration are thus parameterized as follows in the model:

$$^{18}\alpha_{marine} = \frac{1}{1.025} \times \frac{1}{1.003389} = 0.9723$$

We calculated the isotopic effects of terrestrial biological O_2 uptake by subtracting the equilibrium fractionation of O_2 leaf water relative to air from the total terrestrial respiratory fractionation, both of which were again obtained from Table 6 of Luz et al. (2014):[54]

$$^{18}\epsilon_{terrestrial} = (17.7 \pm 1.0) - (0.75 \pm 0.05) = 16.95$$

Terrestrial respiratory oxygen isotope fractionation is amplified due to the enrichment in $\delta'^{18}O$ of leaf water mentioned earlier. The combined isotopic effects of these two processes is thus parameterized as follows in the model:

$$^{18}\alpha_{terrestrial} = \frac{1}{1.01695} \times \frac{1}{1.0045} = 0.9789$$

Oxygen isotope fractionation during biological O_2 uptake imparts its own $\Delta'^{17}O$ signature via slight variations in the mass-dependent fractionation of different biological O_2 uptake pathways.[11] Important biological O_2 uptake pathways include the cytochrome c oxidase pathway (COX), the alternative oxidase pathway (AOX), and photorespiration (PR), all of which fractionate oxygen isotopes differently. Angert et al. (2003) reported $\theta_{COX} = 0.516 \pm 0.001$ and $\theta_{AOX} = 0.514 \pm 0.001$.[15] The oxygen triple-isotope effects of photorespiration were not explicitly reported in Helman et al. (2005). The oxygen triple-isotope effects of O_2 consumption by Rubisco oxygenase and glycolate oxidase, the two pathways by which photorespiration occurs, were reported as γ values which is an alternate formulation for quantifying the oxygen triple-isotope effects of a reaction.[28] γ is defined as follows:

$$\gamma = \frac{1 -^{18} \alpha^{\theta}}{1 -^{18} \alpha}$$

We rearranged this equation to determine the $\theta_{O2\ uptake}$ values of O_2 consumption via Rubisco oxygenase and glycolate oxidase using the $^{18}\varepsilon$ values reported for these processes, 21.3‰ and 21.5‰ respectively. We calculated the $\theta_{O2\ uptake}$ value of photorespiration by assuming that for every two moles of O_2 taken by Rubisco an additional one is consumed by glycolate oxidase, which results in $\theta_{PR} = 0.509 \pm 0.001$.[56]

$\theta_{O2\ uptake}$ measurements for biological O_2 uptake pathways from the Hebrew University group analyzed in the presence of Ar were compromised by pressure-baseline effects and underestimate correct values by about ~0.004.[4] We determined this offset by comparing the Rice University $\theta_{dark\ resp.}$ value of 0.520 ± 0.001 to the Hebrew University θ_{COX} of 0.516 ± 0.001. We note that the θ_{COX} measurement from Hebrew University directly probes the isotopic effects of the cytochrome c oxidase pathway while the $\theta_{O2\ uptake}$ measured at Rice University applies to dark respiration, which occurs through the cytochrome c oxidase and alternative oxidase pathways. $\theta_{dark\ resp.}$ should be directly comparable to θ_{COX} because 1) the cytochrome c oxidase pathway accounts for 90% of dark respiration and 2) the alternative oxidase pathway has a $\theta_{O2\ uptake}$ value extremely close to the cytochrome c oxidase pathway. To determine the global $\theta_{O2\ uptake}$ in the Rice University reference frame, we use $\theta_{dark\ resp.} = 0.520$ as measured in our lab and scale up $\theta_{O2\ uptake}$ values measured for biological O_2 uptake pathways in the presence of Ar at Hebrew University by adding 0.004 to each. This gives us $\theta_{AOX} = 0.518 \pm 0.001$ and $\theta_{PR} = 0.513 \pm 0.002$ for comparable Rice University $\theta_{O2\ uptake}$ values. We weight these $\theta_{O2\ uptake}$ values using the global breakdown of biological O_2 uptake pathways estimated by Blunier et al. (2002), COX = 68%, AOX = 8%, PR = 24%, which gives us a global average $\theta_{O2\ uptake}$ estimate of 0.5182 in the Rice University reference frame and of 0.5142 in the Hebrew University reference frame.[13]

4 Results

4.1 Comparison with target values

We tuned four parameters to have model results match target values obtained from the literature. The first is the global gross oxygen productivity rate, k_p/k_r, which we tuned to match the modern mixing ratio of O_2 in the atmosphere. The second is α_{MIF}, which we tuned for the $\Delta'^{17}O$ composition of stratospheric O_3. The third is the $\delta'^{18}O$ enrichment in global leaf water, which we tuned to be in accord with the $\delta'^{18}O$ of modern tropospheric O_2. The fourth is the transportation normalization factor, which we tuned to the $\Delta'^{17}O$ composition of stratospheric CO_2. The model begins with no O_2 in the atmosphere, and we run it for 100,000 years to ensure that it reaches a steady state in all model results shown in this work. Table 3 shows the steady-state base model outputs compared to target values.

The target mixing ratios of O_2 and CO_2 are 0.21 and 2.7×10^{-4} respectively, which are appropriate for the pre-industrial atmosphere.[57] The modeled mixing ratio of CO_2, 2.7×10^{-4}, and the modeled mixing ratio of O_2, 0.21, are right on target. We use a target mixing ratio of O_3 equal to 2×10^{-6}, which we estimate to be appropriate for the stratosphere at around ~25 km altitude. The model predicts a mixing ratio for ozone of 6.6×10^{-6}. Our model does not include catalytic cycles of NO_x, HO_x, Cl, and Br which drawdown O_3 which might account for the discrepancy.

The target $\delta'^{17}O$, $\delta'^{18}O$ and $\Delta'^{17}O$ values of tropospheric O_2 are 12.105 ± 0.065‰, 23.761 ± 0.104‰ and -0.441 ± 0.012‰ respectively.[6] The modeled $\delta'^{17}O$ of tropospheric O_2, 12.155‰, is within observed uncertainties. The modeled $\delta'^{18}O$ of tropospheric O_2, 23.687‰., is within observed uncertainties. The modeled $\Delta'^{17}O$ of tropospheric O_2, -0.337‰, is ~0.1 ‰ lower than the observed value, which is an offset that we explore in the analysis to follow.

The target $\delta'^{17}O$, $\delta'^{18}O$ and $\Delta'^{17}O$ values of stratospheric O_3 are 77 ± 10‰, 94 ± 13‰ and 29 ± 4‰ respectively.[58] The modeled $\delta'^{17}O$ of stratospheric O_3 is 75‰, which is within observed uncertainties. The modeled $\delta'^{18}O$ of stratospheric O_3 is 87‰, which is within observed uncertainties. The modeled $\Delta'^{17}O$ of stratospheric O_3 is 29‰, which is right on target. We calculated the target stratosphere-troposphere $\Delta'^{17}O$ CO_2 flux (‰ moles CO_2 year^{-1}) using the methodology of Boering et al. (2004) with updated stratospheric CO_2 isotopic data from Wiegel et al. (2013).[49,59] We obtain an estimate of 4.01×10^{15} ‰ moles CO_2 year^{-1} for this flux which is slightly higher than the estimate of 3.6×10^{15} ‰ moles CO_2 year^{-1} calculated using data from Boering et al. (2003). Our modeled flux is 7.36×10^{15} ‰ moles CO_2 year^{-1} which is a little less than double the targeted value. This is the same order of magnitude difference observed in Young et al. (2014).[3]

The target $\delta'^{17}O$, $\delta'^{18}O$, and $\Delta'^{17}O$ values of tropospheric CO_2 were calculated using tropospheric CO_2 isotopic data from Liang et al. (2017) while the target $\delta'^{17}O$, $\delta'^{18}O$ and $\Delta'^{17}O$ values of stratospheric CO_2 were calculated using data from Wiegel et al. (2013).[49,60] Data were divided into 8 equal-height bins and evaluated using the following equation for the $\Delta'^{17}O$ of CO_2:

$$\Delta'^{17}O\ CO_2(\Delta h) = \left(\int_{\Delta h} n(h)dh / \int_{h_1}^{h_2} n(h)dh\right)$$

Analogous equations were used to calculate the target $\delta'^{17}O$ and $\delta'^{18}O$ values. Δh is the bin height, n is number density, and h is height. The target $\delta'^{17}O$, $\delta'^{18}O$ and $\Delta'^{17}O$ values of tropospheric CO_2 are 21.3‰, 40‰ and 0.2‰ respectively. The modeled $\delta'^{17}O$ of tropospheric CO_2, 22.5‰, is ~1.2‰ higher than the target value. The modeled $\delta'^{18}O$ of tropospheric CO_2, 42.5‰, is ~2.5‰ higher than the target value. The modeled $\Delta'^{17}O$ of tropospheric CO_2, 0.04‰, is

~0.2‰ less than the target value. We note that literature estimates for the isotopic composition of tropospheric CO_2 are highly uncertain, so we are not too concerned about the offsets between modeled and target values for this species. The target $\delta'^{17}O$, $\delta'^{18}O$ and $\Delta'^{17}O$ values of stratospheric CO_2 are 22.9‰, 40.4‰, and 1.55‰ respectively. The modeled $\delta'^{17}O$ of stratospheric CO_2, 25.6‰, is ~2.7‰ higher than the target value.The modeled $\delta'^{18}O$ of stratospheric CO_2, 45.6‰, is ~5.2‰ higher than the target value. The modeled $\Delta'^{17}O$ of stratospheric CO_2, 1.534‰, is in-line with the target value.

The high predicted $\delta'^{18}O$ values of stratospheric and tropospheric CO_2 occur because we did not include the isotopic effects of soil respiration on the isotopic composition of atmospheric CO_2. Our model instead accounts for the CO_2 that enters leaves during photosynthesis, equilibrates with leaf water enriched by evapotranspiration, and diffuses back into the atmosphere.[61] Soil water is unaffected by evapotranspiration, meaning that it is depleted relative to the source water used by terrestrial photosynthesis in higher plants. The CO_2 produced by soil respiration equilibrates with this water reservoir and is thus depleted in $\delta'^{18}O$ compared to the CO_2 equilibrated with enriched leaf water during photosynthesis.[62] If we included the isotopic composition of CO_2 produced by soil respiration in our model, the resulting $\delta'^{18}O$ values for stratospheric and tropospheric CO_2 would be lower. Given that we are primarily interested in the $\Delta'^{17}O$ of stratospheric and tropospheric CO_2 we do not parameterize the isotopic effects of this process because it should not change $\Delta'^{17}O$ results significantly. Oxygen isotope exchange between CO_2 and H_2O is characterized by $\theta = 0.528$, the same as the reference slope used to define $\Delta'^{17}O$, which means that the oxygen triple-isotope effects of soil respiration would not impact the modeled $\Delta'^{17}O$ of atmospheric CO_2. Given that we are more interested in the $\Delta'^{17}O$ of atmospheric CO_2 than the absolute value of the $\delta'^{18}O$ of atmospheric CO_2, we are comfortable leaving the modeled $\delta'^{18}O$ of stratospheric and tropospheric CO_2 as they are.

4.2 Impact of terrestrial and marine biosphere productivity on $\Delta'^{17}O$ $O_{2,\,trop}$

Our initial work with the model focused on isolating the source of the 0.05‰ difference between observations of $\Delta'^{17}O$ $O_{2,\,trop}$ in the Rice University and Hebrew University reference frames. After getting the model up and running we also observed a large, ~0.1 ‰ offset between the base model prediction of the oxygen triple-isotope signature of tropospheric O_2, -0.339‰, and the most recent measurement of $\Delta'^{17}O$ $O_{2,\,trop}$, -0.441 ± 0.012‰.[6] Given that this model run was conducted using parameters from the same laboratory reference frame as the air observation we compared it to, we found it strange that the two did not match one another. This offset suggested that our model was missing an important component of the oxygen triple-isotope budget of tropospheric O_2. We hypothesized that the relative contributions of terrestrial versus marine productivity to global gross oxygen productivity could be a significant control on the $\Delta'^{17}O$ of tropospheric O_2 because terrestrial and marine O_2 production and consumption processes have distinct isotopic effects. To explore this idea we split the global biosphere/hydrosphere inherited from the original model into terrestrial and marine components to account for their distinct source water oxygen isotope compositions and differing oxygen isotope fractionation processes.[45,55]

Figure 6 shows how changing the proportion of global gross oxygen productivity coming from the terrestrial biosphere versus the marine biosphere impacts $\Delta'^{17}O$ $O_{2,\,trop}$ for a fixed global gross oxygen productivity. We plot observations of $\Delta'^{17}O$ $O_{2,\,trop}$ from Wostbrock et al. (2020) and Luz and Barkan (2011) alongside model results.[6,32] The Wostbrock et al. (2020) observation of $\Delta'^{17}O$ $O_{2,\,trop}$ is directly comparable with model results using $\theta_{O2\ uptake}$ values in the Rice University reference frame because both laboratories strip Ar from O_2 prior to isotopic analysis and correct for pressure-baseline effects in similar ways. We used VSMOW vs. air oxygen isotope data from Luz and Barkan (2011) to calculate the $\Delta'^{17}O$ $O_{2,\,trop}$ measured at Hebrew University.[32] The Hebrew University observation of $\Delta'^{17}O$ $O_{2,\,trop}$ is directly comparable to model results using Hebrew University $\theta_{O2\ uptake}$ measurements because they come from the same laboratory.

We find that $\Delta'^{17}O$ $O_{2,\,trop}$ increases by ~0.07‰ when the proportion of global gross oxygen productivity coming from the terrestrial biosphere rises from 0 to 100% in the Rice University model run, and that $\Delta'^{17}O$ $O_{2,\,trop}$ increases by ~0.1‰ when the proportion of global gross oxygen productivity coming from the terrestrial biosphere rises from 0 to 100% in the Hebrew University model run. There is no relative proportion of terrestrial versus marine gross oxygen productivity where either set of model results are consistent with the $\Delta'^{17}O$ $O_{2,\,trop}$ measured in their respective reference frames. This indicates that the inclusion of terrestrial and marine biospheres does not help solve the disagreement between model results and observations in the two laboratory reference frames we are interested in.

4.3 Biological O_2 uptake is an important control on $\Delta'^{17}O$ $O_{2,\,trop}$

We next wanted to quantify the effects of different O_2 consumption processes on the oxygen triple-isotope composition of tropospheric O_2. Figure 7 shows model results when all of global O_2 consumption occurs through one O_2 consumption process, i.e. global $\theta_{O2\ uptake}$ is set equal to the $\theta_{O2\ uptake}$ value of different biological O_2 uptake pathways. The $\theta_{O2\ uptake}$ for photorespiration, 0.509 ± 0.002, is from Helman et al. (2005).[28] The oxygen triple-isotope effects of the terrestrial Mehler reaction and marine Mehler-like reactions were reported as γ values by Helman et al. (2005), which when converted to θ values lead to $\theta_{MRt} = 0.525 \pm 0.002$ and $\theta_{MRm} = 0.496 \pm 0.004$. We consider the $\theta_{O2\ uptake}$ of the Mehler reaction measured in spinach thylakoids to be representative of the terrestrial Mehler reaction and the $\theta_{O2\ uptake}$ measured on Mehler-like reactions in *Synechocystis* to be representative of marine Mehler-like reactions.[28] The $\theta_{O2\ uptake}$ for the alternative oxidase pathway, 0.514 ± 0.001, and the $\theta_{O2\ uptake}$ for the cytochrome c oxidase pathway, 0.516 ± 0.001, are from Angert et al. (2003).[15] The $\theta_{O2\ uptake}$ for dark respiration, 0.520 ± 0.001, is from Ash et al. (2020) and is the only $\theta_{O2\ uptake}$ value included in this analysis which was measured directly at Rice University.[9]

Higher global $\theta_{O2\ uptake}$ values lead to higher predicted $\Delta'^{17}O\ O_{2,\ trop}$ values while lower global $\theta_{O2\ uptake}$ values lead to lower predicted $\Delta'^{17}O\ O_{2,\ trop}$ values. The modeled $\Delta'^{17}O\ O_{2,\ trop}$ values that result from using these end-member $\theta_{O2\ uptake}$ values for global $\theta_{O2\ uptake}$ span ~0.9 ‰ in $\Delta'^{17}O$ space and range from -0.167‰ to -0.871‰. This is a huge variation when compared to the marginal impact of varying the proportion of global gross oxygen productivity coming from the terrestrial and marine biospheres. When $\theta_{O2\ uptake}$ is set equal to 0.516, the θ_{COX} value measured at Hebrew University, the predicted $\Delta'^{17}O\ O_{2,\ trop}$ is -0.410‰. When $\theta_{O2\ uptake}$ is set equal to 0.520, the $\theta_{dark\ resp.}$ value measured at Rice University, the predicted $\Delta'^{17}O\ O_{2,\ trop}$ is -0.312‰. This large 0.1‰ offset in predicted $\Delta'^{17}O\ O_{2,\ trop}$ values using $\theta_{O2\ uptake}$ values characterizing the same process suggests that methodological differences between the two laboratories in the measurement of $\theta_{O2\ uptake}$ values characterizing biological O_2 uptake processes likely play a large role in disparate model predictions of $\Delta'^{17}O\ O_{2,\ trop}$ and in interlaboratory disagreements on the oxygen-triple isotope composition of tropospheric O_2.

4.4 Varying fluxes of O_2 consumed by dark respiration and photorespiration isn't sufficient

We next wanted to explore how the global expression of different biological O_2 consumption processes impacts the oxygen triple-isotope composition of tropospheric O_2. Figure 8 shows model simulations assuming that the cytochrome c oxidase pathway and photorespiration are the only two O_2 consumption processes contributing to global biological O_2 uptake. In these runs, the global $\theta_{O2\ uptake}$ is the weighted average of these two pathways:

$$global\ \theta_{O_2\ uptake} = f_{COX}\theta_{COX} + f_{PR}\theta_{PR}$$

Where f_{COX} is the fraction of the global O_2 consumption flux taken by the cytochrome c oxidase pathway and f_{PR} is the fraction of the global O_2 consumption flux taken by photorespiration. Past estimates of the value of global $\theta_{O2\ uptake}$ included the isotopic effects of the cytochrome c oxidase pathway, photorespiration, and the alternative oxidase pathway. In this set of model simulations we ignore the isotopic effects of alternative oxidase pathway because the observed value of θ_{AOX}, 0.514 ± 0.001, is in between θ_{COX}, 0.516 ± 0.001, and θ_{PR}, 0.509 ± 0.002. This means that including the alternative oxidase pathway as an end-member in these calculations would not impact the calculated global $\theta_{O2\ uptake}$ in a different manner than the cytochrome c oxidase pathway and photorespiration do.

The cytochrome c oxidase pathway accounting for 10% of global O_2 consumption and photorespiration accounting for 90% of global O_2 consumption reconciles the Rice University model run with the $\Delta'^{17}O$ $O_{2,\ trop}$ observation in the Rice University reference frame. Given that it is well established that the cytochrome c oxidase pathway is the dominant O_2 consumption globally, we consider this situation to be unfeasible in reality. A 50/50 global breakdown of the cytochrome c oxidase pathway versus photorespiration in the global O_2 consumption budget reconciles the Hebrew University model run with the $\Delta'^{17}O$ $O_{2,\ trop}$ observation in the Hebrew University reference frame. This situation is plausible, as it is not impossible that more photorespiration is occurring in the biosphere than previously understood. However, the fact that the Rice University and Hebrew University model runs predict different contributions of the cytochrome c oxidase pathway and photorespiration to the global O_2 consumption budget suggests that we are missing a key part of the global O_2 budget, because in principle the same global relative expression of biological O_2 consumption pathways should explain both laboratory offsets. Including a biological O_2 uptake pathway with a $\theta_{O2\ uptake}$ value lower than photorespiration in the global O_2 consumption budget could help reconcile model results with observations in both reference frames by decreasing predicted $\Delta'^{17}O$ $O_{2,\ trop}$ values. The only O_2 consumption processes with a $\theta_{O2\ uptake}$ value lower than photorespiration are marine Mehler-like reactions, which are characterized by a $\theta_{O2\ uptake}$ value of 0.496 ± 0.004.

4.5 Inclusion of Mehler-like reactions leads to consistency in two laboratory scales

Figure 9 shows predicted $\Delta'^{17}O$ $O_{2,\ trop}$ values as the O_2 consumption flux of marine Mehler-like reactions increases from 0 to 50% of marine gross oxygen productivity. In these runs the global $\theta_{O2\ uptake}$ is the weighted average of all the biological oxygen uptake pathways we have discussed in this work:

$$global\theta_{O_2\ uptake} = f_{COX}\theta_{COX} + f_{PR}\theta_{PR} + f_{AOX}\theta_{AOX} + f_{MRt}\theta_{MRt} + f_{MRm}\theta_{MRm}$$

Where $f_{COX.}$ is the fraction of the global O_2 consumption flux taken by the cytochrome c oxidase pathway, f_{PR} is the fraction of the global O_2 consumption flux taken by photorespiration, f_{AOX} is the fraction of the global O_2 consumption flux taken by the alternative oxidase pathway, f_{MRt} is the fraction of the global O_2 consumption flux taken by the terrestrial Mehler reaction, and f_{MRm} is the fraction of the global O_2 consumption flux taken by marine Mehler-like reactions. Rice University $\theta_{O2\ uptake}$ values for the terrestrial Mehler reaction and marine Mehler-like reactions are the same as their Hebrew University counterparts because Ar was stripped from O_2 prior isotopic analysis for these two processes which means that there should not be interlaboratory differences in their measured oxygen triple-isotope effects.[28] $\theta_{O2\ uptake}$ values for all the biological O_2 consumption processes used in this work can be found in Table 4.

In these model runs we assume that the flux of O_2 consumed by the terrestrial Mehler reaction is equal to 10% of terrestrial gross oxygen productivity and that the flux of O_2 through the alternative oxidase pathway flux is equal to 10% of the flux of O_2 through the cytochrome c oxidase pathway.[2,25] We adopt a terrestrial gross carbon productivity of 9.4×10^{15} mol C year^{-1} with 72.5% of gross carbon productivity coming from C_3 plants and the remaining 27.5% coming from C_4 plants.[63, 64] We apply a photosynthetic quotient factor of 1.07 mol O_2 / mol C to convert this flux into units of O_2.[65] We assume that C_4 plants do not undergo photorespiration, and calculate the fraction of gross oxygen productivity produced by C_3 plants that is consumed by dark respiration and photorespiration using the following relationship from vonCaemmerer and Farquhar (1981):[66]

$$\frac{dark\,resp. + photoresp.}{dark\,resp.} = \left(\frac{4.5}{4}\right)\left(\frac{CO_2 \times (p_i/p_a) + (7/3) \times \Gamma_*}{CO_2 \times (p_i/p_a) - \Gamma_*}\right)$$

Where Γ_* is the CO_2 compensation point, 34 ppm, p_i/p_a is the ratio of CO_2 inside the leaf to the atmospheric value, 0.65, and p_a is the pre-anthropogenic atmospheric CO_2 concentration, 281 ppm. We divide the total terrestrial gross oxygen productivity flux by 0.9 to account for the extra O_2 consumed by the terrestrial Mehler reaction.

We assume that photorespiration does not occur in the ocean. We adopt a marine net carbon productivity of 4.04×10^{15} mol C / yr, and scale this ^{14}C-based estimate to be in units of O_2 by taking 0.37 to be the ratio of net carbon productivity to gross oxygen productivitiy.[67,13] All of this marine O_2 production is consumed by the cytochrome c oxidase pathway and the alternative oxidase pathway. We calculate the O_2 uptake flux of marine Mehler-like reactions by positing that it consumes an additional amount of O_2 equal to between 0 and 50% of total marine gross oxygen productivity. From these calculations we obtain global fluxes of O_2 consumed by the cytochrome c oxidase pathway, the alternative oxidase pathway, photorespiration, terrestrial

Mehler reaction, and marine Mehler-like reactions. We then divide each of these fluxes by the global O_2 consumption flux to determine the fraction of global O_2 consumption which occurs via each of the pathways. We calculate global $\theta_{O2\ uptake}$ values for each model run by taking a weighted average of the process-specific $\theta_{O2\ uptake}$ values as shown in the first equation in this section.

As the flux of O_2 consumed by marine Mehler-like reactions increases from 0 to between 40 and 50% of marine gross oxygen productivity, both Rice University and Hebrew University model results converge on the $\Delta'^{17}O\ O_{2,\ trop}$ observations in their respective reference frames. The fact that the same flux of O_2 being consumed by marine Mehler-like reactions reconciles model results with observations in two laboratory reference frames with differing methods of O_2 isotopic analysis indicates that marine Mehler-like reactions play a large, previously unrecognized role in global biological O_2 uptake, and that the ~0.1‰ offset between the Rice University and Hebrew University observations of $\Delta'^{17}O\ O_{2,\ trop}$ is purely an issue of mismatched laboratory reference frames.

5 Discussion

5.1 Mehler-like reactions help resolve Δ'^{17}O-GOP/^{14}C-PP discrepancies

Figure 10 shows concurrently measured Δ'^{17}O-based gross oxygen productivity rates versus ^{14}C-based net carbon productivity rates from various locations in the ocean. We include observations from the Western Equatorial Pacific (WEP), the Central Equatorial Pacific (CEP), the Southern Ocean (SO), and the Bermuda Atlantic Time-series (BATS), Hawaii Ocean Time-series (HOT), and California Cooperative Oceanic Fisheries Investigations (CalCOFI) sites.[68–72] We plot the ratio of gross oxygen productivity to net carbon productivity, 3.3, observed in experiments conducted on the green alga *Dunaliella tertiolecta* in Halsey et al. (2010).[37] We take this observation to represent the ratio of gross oxygen productivity to net carbon productivity produced in organisms that utilize the terrestrial Mehler reaction. We suggest that O_2 consumption by marine Mehler-like reactions was not observed in the experiments conducted on *Dunaliella tertiolecta* because this species uses O_2 photoreduction pathways similar to those that occur in terrestrial Mehler reaction.[73] Therefore, we also plot the result of multiplying this ratio by a factor of 1.45 to approximate the effect that marine Mehler-like reactions consuming ~45% of marine gross oxygen productivity would have in biasing $\Delta'^{17}O$-based gross O_2 production estimates upwards. This bias occurs because the isotopic composition of photosynthetic O_2 is imparted in dissolved O_2 regardless of whether the O_2 is later consumed, meaning that gross oxygen productivity rates determined using the oxygen triple-isotope mass-balance approach trace all O_2 evolved during cellular processes, not just the O_2 used to fix carbon into biomass.

We find that the original ratio of gross oxygen productivity to net carbon productivity observed in Halsey et al. (2010) can account for $\Delta'^{17}O$-gross oxygen productivity/^{14}C-net carbon productivity discrepancies in the Central Equatorial Pacific and at the Hawaii Ocean Time-series site, while the ratio that accounts for a large contribution of marine Mehler-like reactions to marine biological O_2 uptake can account for $\Delta'^{17}O$-gross oxygen productivity/^{14}C-net carbon productivity discrepancies in the Central Equatorial Pacific, the Southern Ocean, and at the CalCOFI and Bermuda Atlantic Time-series sites. While the uncertainties of the Western Equatorial Pacific data do not completely overlap with the slope including the isotopic effects of marine Mehler-like reactions, qualitatively these data are better explained by this relationship as well. These results indicate that a significant portion of the difference between $\Delta'^{17}O$-GOP and ^{14}C-NCP can be ascribed to actual, physiological differences in O_2 and C cycling during photosynthesis caused by the activity of marine Mehler-like reactions.

5.2 Could variations in $\Delta'^{17}O$ $O_{2,\,trop}$ since the LGM reflect O_2 consumption?

Previous work used the oxygen triple-isotope composition of atmospheric O_2 trapped in bubbles stored in glacial ice to estimate variations in global gross oxygen productivity (i.e. terrestrial + marine productivity) over the last 400,000 years.[2,13] The $\Delta'^{17}O$ of O_2 stored in glacial ice records the isotopic budget of the troposphere at the time it was trapped because the atmosphere is well-mixed with respect to O_2.[53] The $\Delta'^{17}O$ of atmospheric O_2 reflects the balance of stratospheric photochemistry and biological O_2 production. When normalized to the co-variation of $\delta'^{17}O$ and $\delta'^{18}O$ through biological O_2 uptake, this signal allows one to estimate rates of global biosphere gross oxygen productivity on glacial-interglacial timescales.

Figure 11 shows variations in the $\Delta'^{17}O$ of atmospheric O_2 obtained from the GISP-2 glacier site since the Last Glacial Maximum (LGM) as reported in Blunier et al. (2012).[2] The $\Delta'^{17}O$ of modern tropospheric O_2 and of tropospheric O_2 at the LGM are plotted alongside these data. Blunier et al. (2012) inferred the ~0.03‰ decrease in $\Delta'^{17}O$ $O_{2,\,trop}$ to mean that global gross oxygen productivity at the LGM was around 90% of that at present. Their estimate did not account for any O_2 consumption occurring through marine Mehler-like reactions. This is problematic because for every ~10% increase in the fraction of marine gross oxygen productivity consumed by marine Mehler-like reactions $\Delta'^{17}O$ $O_{2,\,trop}$ decreases by ~0.01-0.02‰, which is comparable to the total variation in $\Delta'^{17}O$ $O_{2,\,trop}$ since the LGM. How might the quantity of O_2 consumed by marine Mehler-like reactions have changed since the LGM, and what could this imply for global gross oxygen productivity estimates made using $\Delta'^{17}O$ $O_{2,\,trop}$ over this time period?

Global average sea-surface temperatures have risen by around 3.1 °C since the LGM while tropical average sea-surface temperature have risen by around 3.5 °C.[74] Low temperatures appear to lead to higher abundances of the A-type flavodiiron proteins associated with marine Mehler-like reactions in *Synechococcus,* the second-most dominant cyanobacteria in the global ocean.[75,76] Figure 13 shows the results of experiments conducted on temperate and tropical strains of *Synechococcus* in which rates of O_2 consumption by Mehler-like reactions were compared in cultivations conducted at 18 °C and 22 °C. Upregulation of marine Mehler-like reactions at lower temperatures is stronger in the strain of *Synechococcus* from the tropics (i.e. low latitudes, higher temperatures) than the strain from temperate latitudes (i.e. lower temperatures). These observations indicate that marine Mehler-like reactions were likely upregulated under the cooler surface conditions of the LGM, especially in the tropical ocean. Removing the isotopic effects of Mehler-like reactions would cause the $\Delta'^{17}O$ $O_{2,\ trop}$ at the LGM to be higher than what is observed in the record, because including marine Mehler-like reactions in the global O_2 budget lowers $\Delta'^{17}O$ $O_{2,\ trop}$. This suggests that there could have been a larger decrease in $\Delta'^{17}O$ $O_{2,\ trop}$ since the LGM that can be ascribed to the isotopic effects of global gross oxygen productivity, which indicates that global gross oxygen productivity was lower at the LGM than was estimated by Blunier et al. (2012).

Atmospheric dust fluxes were much larger in the LGM than at present due to the drier climate predominant at the time.[77] Continentally-derived dust stimulates marine primary productivity through the deposition of iron, a key limiting nutrient on phytoplankton growth, into the ocean.[78] Fe is also a key structural component of the flavodiiron proteins which mediate the photoreduction of O_2 to H_2O through marine Mehler-like reactions.[79] Dust-mediated, soluble Fe deposition fluxes are estimated to have been two times as large at the LGM as in the pre-industrial climate, 0.20-0.82 Tg/yr versus 0.08-0.38 Tg/yr.[80] If marine Mehler-like reactions are Fe-limited in parts of the ocean due to their dependence on flavodiiron proteins, this suggests that the potential for marine Mehler-like reactions activity at the LGM could have been higher than in the modern. Using similar logic to that laid out in the case of the increase in temperature since the LGM discussed in the previous paragraph, this suggests that global gross oxygen productivity at the LGM was lower than that predicted by previous $\Delta'^{17}O$ estimates which did not account for the isotopic effects of marine Mehler-like reactions.

While we are not in the position to quantitatively determine the degree to which changes in O_2 consumption fluxes contributed to the variation in $\Delta'^{17}O_{2,\ trop}$ observed since the LGM, a qualitative understanding of the processes impacting Mehler-like reaction activity suggests that this large lever on $\Delta'^{17}O_{2,\ trop}$ changed in global importance since the LGM. This indicates that variations in O_2 consumption processes need to be accounted for when making estimates of global gross oxygen productivity using the oxygen-triple isotope system.

5.3 Proposed experiments

Our hypothesis that marine Mehler-like reactions play an important role in the global O_2 consumption relies on the observation that the $\theta_{O2\ uptake}$ of marine Mehler-like reactions, 0.496 ± 0.004, is far lower than the $\theta_{O2\ uptake}$ of the other O_2 consumption pathways. Biochemically, it is unclear how a θ value this low could occur in nature. The $^{18}\varepsilon$ measured for marine Mehler-like reactions in Helman et al. (2005) was -9.6‰, which the authors note is around 5% smaller than the observation for the terrestrial Mehler reaction of -15.1‰ observed in Guy et al. (1993).[28,81] Guy et al. (1993) added an artificial electron acceptor, methylamine, which might have down-regulated light-dependent reactions and lead to the observed $^{18}\varepsilon$ in this study being biased upwards by other O_2 consumption processes like dark respiration. Helman et al. (2005) also determined the $\theta_{O2\ uptake}$ and $^{18}\varepsilon$ of the terrestrial Mehler reaction in spinach thylakoids to be 0.525 ± 0.002 and -10.8‰ respectively. It is unclear why the oxygen triple-isotope effects of the terrestrial Mehler reaction in spinach thylakoids and of marine Mehler-like reactions in *Synechocystis* are so radically different. The uncertainties about the isotopic effects of these reactions demand new measurements of the $\theta_{O2\ uptake}$ of Mehler-like reactions to set this cornerstone of our argument on a firm footing.

We propose to measure the oxygen triple-isotope and clumped oxygen isotope effects of photosynthesis and marine Mehler-like reactions in *Synechococcus* to substantiate our hypothesis that marine Mehler-like reactions constitute a globally important O_2 consumption flux. The clumped isotope signature of O_2 quantifies the degree to which the rare isotopes of oxygen, ^{17}O and ^{18}O, prefer to bond with one another in an O_2 molecule relative to what would be expected from a stochastic bonding process. O_2 production and consumption processes impart distinct clumped oxygen isotope effects depending on the degree to which they break or preserve the O-O bond, which in principle would allow clumped isotopes to be used as an independent isotopic check on results from the oxygen triple-isotope system.[9,82] Previous studies of the isotopic effects of marine Mehler-like reactions were conducted on *Synechocystis*, a freshwater cyanobacterium species. *Synechococcus* is a globally widespread marine cyanobacteria whose metabolic pathways are likely more representative of those that occur in marine cyanobacteria than those present in the freshwater species *Synechocystis.* We will conduct our sets of experiments on modified strains of *Synechococcus elongatus*, a fast-growing cyanobacterium that has been widely used as a model organism for metabolic engineering and research into circadian rhythms.[83,84]

The first of our proposed experiments is a re-determination of the isotopic effects of photosynthesis observed in Helman et al. (2005).[28] This experiment will be conducted on a mutant strain of *Synechococcus elongatus* defective in the terminal oxidases that enable dark respiration and lacking the the A-type flavodiiron proteins (Δ*flv1*/Δ*flv3*) which facilitate the activity of marine Mehler-like reactions.[85] We will grow *Synechococcus* at ambient CO_2 levels because rates of photorespiration are low in cyanobacteria due to carbon concentrating mechanisms.[86] With the removal of dark respiration, marine Mehler-like reactions, and photorespiration as O_2 consumption pathways, the isotopic signature of O_2 observed in this experiment will solely reflect the oxygen triple-isotope and clumped oxygen isotope effects of photosynthesis. We will run this experiment at a light intensity of 1000 μmol photons m^{-2} s^{-1} which is appropriate for surface ocean conditions in the tropics (between 30°N and 30°S). The O_2 produced during this experimental run will then be collected and the oxygen triple-isotope and clumped oxygen isotope effects of photosynthesis will be determined from measurements conducted at Rice University.

We will determine the oxygen triple-isotope and clumped oxygen isotope effects of marine Mehler-like reactions using methods similar to those used in Helman et al. (2005).[85] This will involve two sets of parallel experiments. One set of parallel experiments will be conducted in the presence of 20 mM methylamine as was done in Guy et al. (1993), while the second set of parallel experiments will be conducted without methylamine present as was done in Helman et al. (2005).[28,81] This will allow us to definitively show why the $^{18}\varepsilon$ values observed for O_2 photoreduction in the two studies were different by ~5‰, with our new experiments capturing the complete isotopic effects of Mehler-like reactions. These two sets of parallel experiments will be conducted on a mutant strain of *Synechococcus elongatus* defective in the genes encoding for the terminal oxidases that power, but with the A-type flavodiiron proteins that enable Mehler-like reactions activated. This means that photosynthesis and Mehler-like reactions will be the only two processes occurring during the runs. Experiments will be conducted at a light intensity of 1000 μmol photons m^{-2} s^{-1}, appropriate for surface ocean conditions in the tropics between 30°N and 30°S, which will help to quantify the isotopic effects of Mehler-like reactions under environmentally realistic conditions.

Given that we will have determined the isotopic effects of photosynthesis in the first proposed experiment, we will be able to use a mass-balance approach to determine the isotopic effects of Mehler-like reactions if we know the relative rates of O_2 production and O_2 consumption during the experiments. The basic mass-balance equation for these experiments is as follows:

$$\frac{d[O_2]}{dt} = GOP - U = NOP$$

Where GOP is gross oxygen productivity, U is the O_2 uptake rate, and NOP is net oxygen productivity. Net oxygen productivity will be determined directly by observing the decrease in the concentration of O_2 in both experiments. The source water for one of the parallel experiments will be spiked with a known amount of $H_2{}^{18,}O$ which will allow the $\delta'^{18}O$ of water measured at the end of this experiment to be used to determine the rate of gross oxygen productivity. The ratio of gross oxygen productivity to net oxygen productivity will be calculated in the spiked experiment, which will allow the gross oxygen productivity in the non-spiked experiment to be determined using observed net oxygen productivity rate. With the GOP and NOP terms accounted for in the above equation, we will be able to calculate the $^{17}\varepsilon$, $^{18}\varepsilon$, and $\theta_{O2\ uptake}$ value of marine Mehler-like reactions in *Synechococcus*, the only consumption process at play in these experiments. If the $\theta_{O2\ uptake}$ value observed for marine Mehler-like reactions is significantly higher than the value of 0.496 ± 0.004 observed in Helman et al. (2005), then our isotopic mass-balance argument in favor of marine Mehler-like reactions as an important global O_2 consumption flux would be disproved.

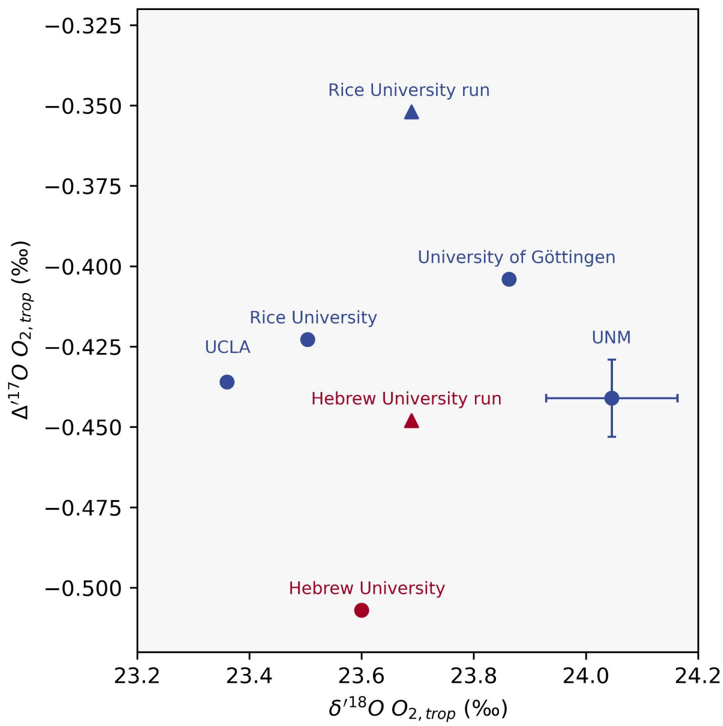
The last set of experiments we propose to conduct will involve observing the response of $\Delta'^{17}O$ produced by *Synechococcus* at increasing light intensities, which will show the importance of light-dependent oxygen uptake in the oxygen triple-isotope system. As light intensity increases, the amount of O_2 consumed by marine Mehler-like reactions will increase as cells protect themselves from the increasing amount of incoming radiation. Figure 12 shows the relationship between the consumption of O_2 by Mehler-like reactions and light intensity in a wild-type strain of *Synechocystis* determined using the $H_2{}^{18}O$ spike-membrane inlet mass spectrometry method in Santana-Sanchez et al. (2019).[21,87] The key observation is that the maximum flux of O_2 consumed by marine Mehler-like reactions roughly doubles, from 41.2 ± 6.0 to 96.4 ± 9.1 μmol O_2 mgChl^{-1} hr^{-1}, as light intensity increases from 500 to 1500 μmol photons m^{-2} s^{-1}. In our experiments, we will grow a wild type strain of *Synechococcus elongatus*, i.e. with photosynthesis, dark respiration, and light-dependent oxygen consumption processes active, under the following light intensities: 10, 20, 50, 100, 200, 500, 1000, and 2000 μmol photons m^{-2} s^{-1}. The $\Delta'^{17}O$ of O_2 produced in these experiments should decrease relative to the oxygen triple-isotope composition of water as light intensity increases. This is because marine Mehler-like reactions, with their low $\theta_{O2\ uptake}$, will exert increasing control on the observed $\Delta'^{17}O$ of O_2. Experimentally proving the importance of light-dependent oxygen uptake on the $\Delta'^{17}O$ of O_2 produced by marine cyanobacteria will show the importance of these reactions for the global oxygen triple-isotope budget of tropospheric O_2. The clumped oxygen isotope effects of photosynthesis and marine Mehler-like reactions determined in the earlier experiments should also show the isotopic effects in clumped oxygen isotope space of increasing fluxes of marine Mehler-like reactions

6 Conclusion

We created a chemical reaction network box model of the $\Delta'^{17}O$ budget of tropospheric O_2. In our model formulation, the biosphere is composed of marine and terrestrial components. Stratospheric parameters were benchmarked to literature estimates for the isotopic composition of relevant chemical species. We identified the oxygen triple-isotope effects of biological O_2 uptake as the most important control on the $\Delta'^{17}O$ of tropospheric O_2. We found that including marine Mehler-like reactions with an O_2 consumption flux equal to between 40 and 50% of marine gross oxygen productivity in the global O_2 budget solves three separate problems at once: 1) interlaboratory disagreements on the isotopic composition of tropospheric O_2, 2) the incompatibility of model predictions of $\Delta'^{17}O$ $O_{2,\ trop}$ using parameters from two laboratory reference frames with comparable air observations, and 3) the significant discrepancies between observed $\Delta'^{17}O$-gross oxygen productivity and ^{14}C-net carbon productivity measurements of marine primary productivity. After having come to this new understanding of the importance of O_2 consumption fluxes in the modern oxygen triple-isotope budget of tropospheric O_2 we suggested that the $\Delta'^{17}O$ variations of O_2 stored in glacial ice since the Last Glacial Maximum may actually also reflect changes in the extent of O_2 consumption processes at the planetary scale rather than solely fluctuations in global gross oxygen productivity. We proposed a series of experiments to test our hypothesis of a large flux of O_2 being consumed by marine Mehler-like reactions which has not been accounted for in past global O_2 budgets. The key takeaway from this research is that O_2 consumption processes play an extremely important role in the $\Delta'^{17}O$ budget of tropospheric O_2 and need to be considered in any application of the oxygen triple-isotope system as a proxy of biological productivity.

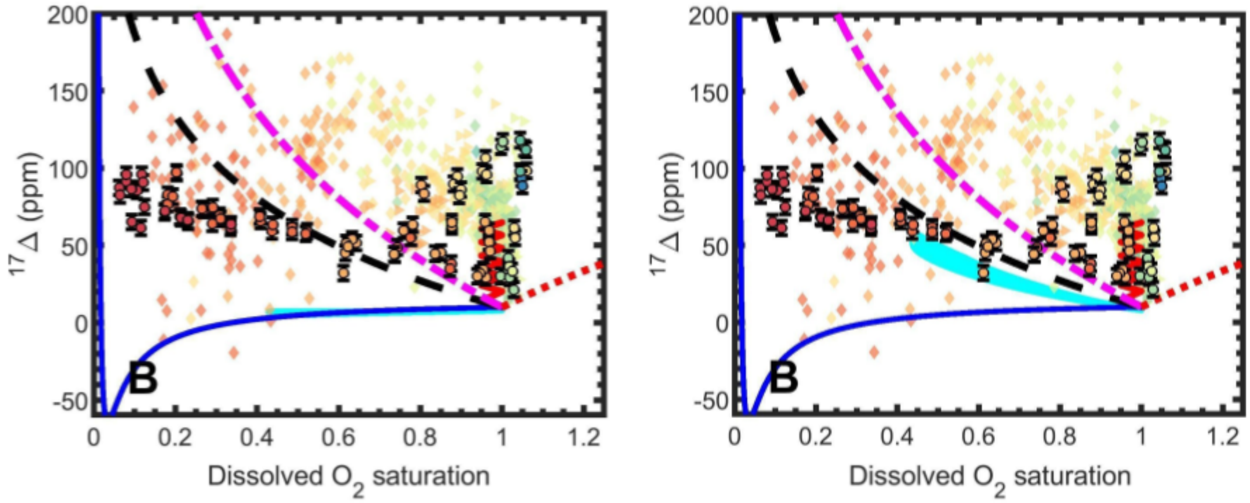
Figures

Figure 1



Caption: $\Delta'^{17}O$ and $\delta'^{18}O$ values of tropospheric O_2 measured at various laboratories. The blue circles are observations made using methodologies similar to those in use at Rice University and the red circle is the observation made at Hebrew University.[32,3,33,5,6] We consider the observation from UNM to be representative of the other observations in blue. The blue triangle is the model result using parameters in the Rice University reference frame and the red triangle is the model result using parameters in the Hebrew University reference frame.

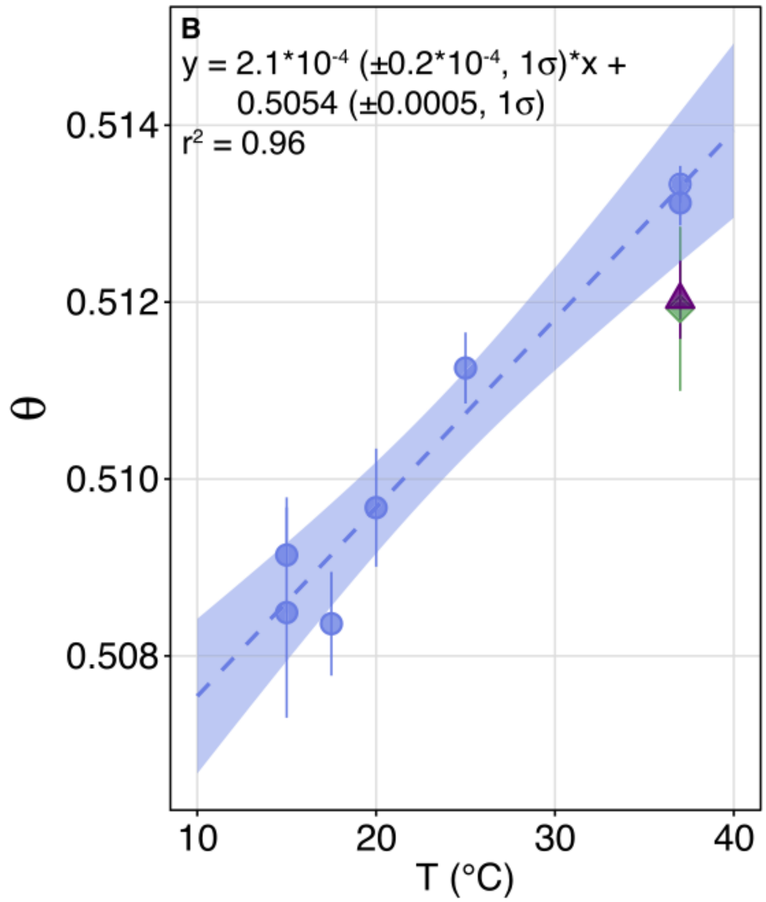
Figure 2



$\theta_{dark\ resp.} = 0.516$ $\theta_{dark\ resp.} = 0.516$

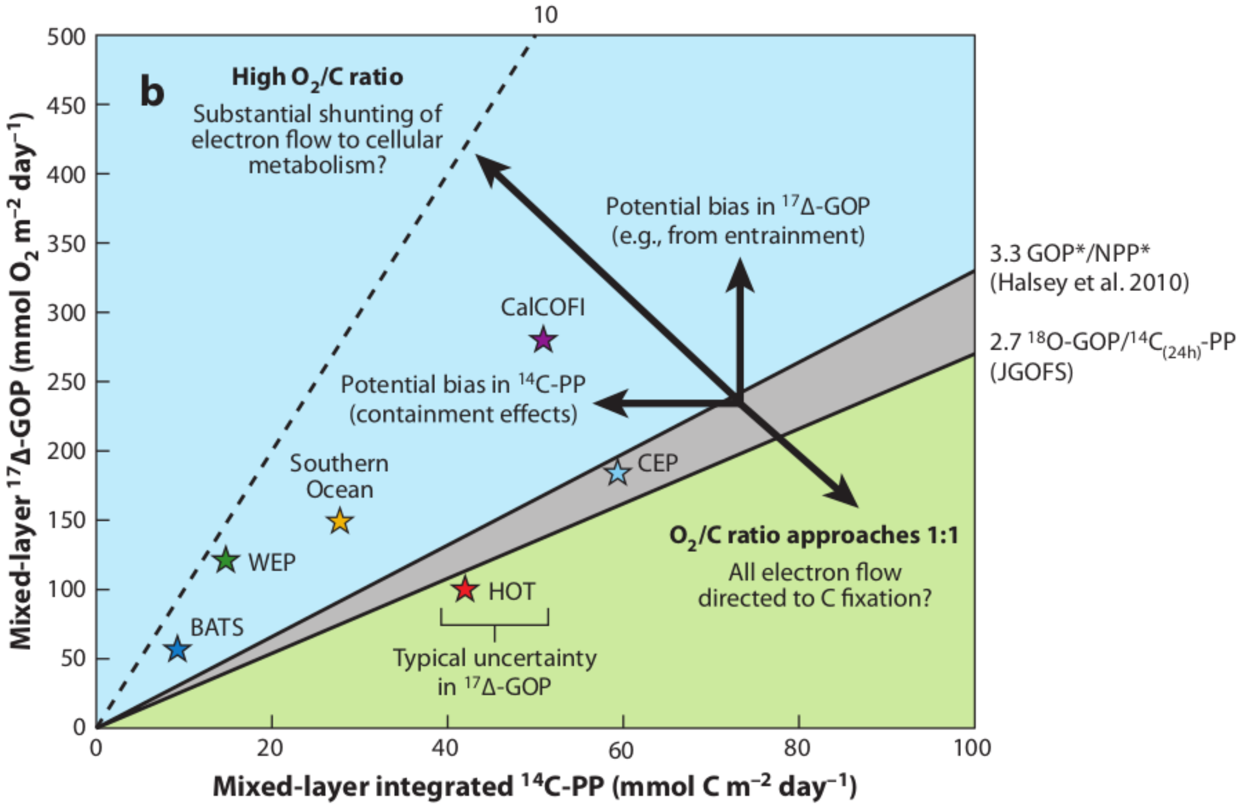
Caption: Plots showing model results plotted against deep ocean oxygen triple-isotope data from upcoming research from our group at Rice. The definition of the oxygen triple-isotope composition of O_2 used in these plots is that used by the oceanographic community and is not the same as the definition used in the rest of the work. Empirical measurements are the circles with error bars. The cyan shaded areas are the model results to focus on. The cyan results on the left were calculated using the $\theta_{dark\ resp.}$ value of 0.516 while cyan results on the right were calculated using the $\theta_{dark\ resp.}$ value of 0.520. The model results using the higher $\theta_{dark\ resp.}$ match deep-ocean oxygen triple-isotope data dissolved O_2 saturation decreases more accurately than the model results using the lower $\theta_{dark\ resp.}$.

Figure 3



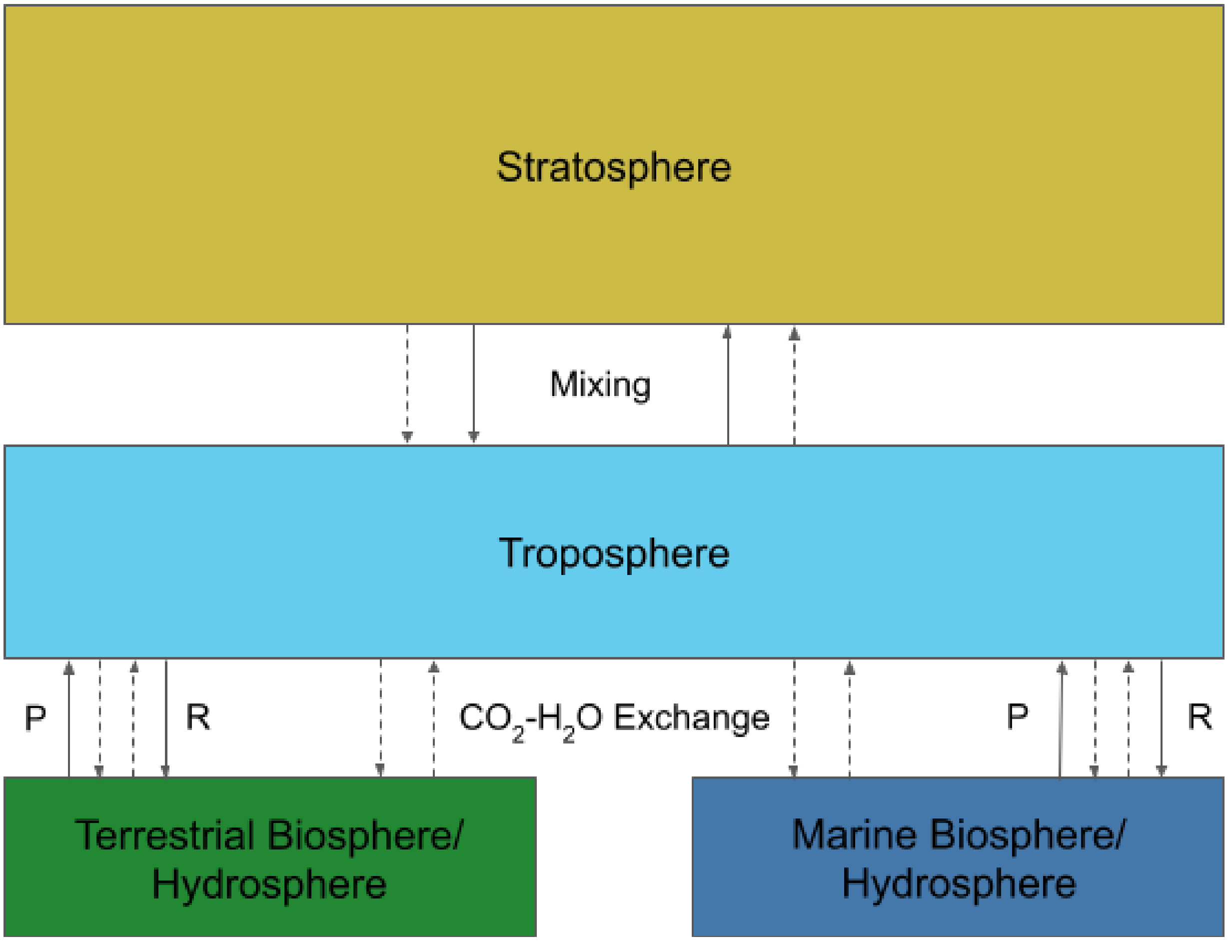
Caption: Plot showing the temperature-dependence of $\theta_{\text{dark resp.}}$ in *E. coli* as found in Stolper et al. (2018).[35]

Figure 4



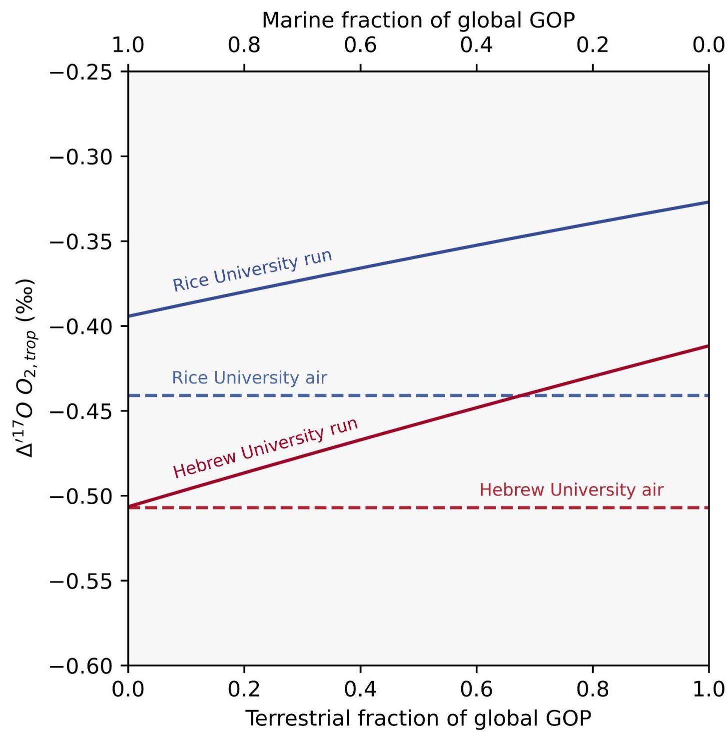
Caption: Plot from Juranek et al. (2013) showing the factors that potentially contribute to observed discrepancies between concurrently measured Δ'^{17}O-gross oxygen productivity and ^{14}C-net carbon productivity rates.[1]

Figure 5



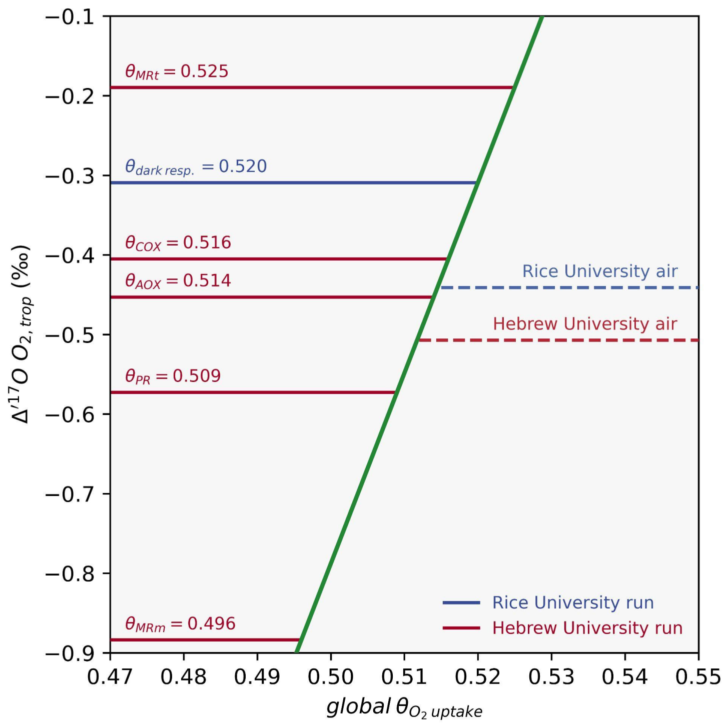
Caption: Schematic of the box model used in this study. Solid arrows represent O_2 fluxes, dashed arrows represent CO_2 fluxes. P is photosynthesis and R is respiration.

Figure 6



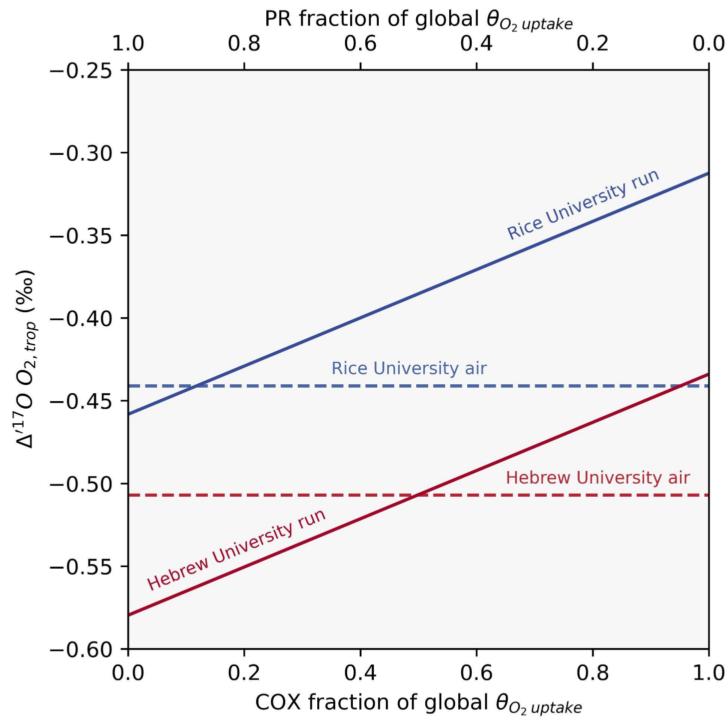
Caption: Plot showing the effect of changing the fraction of global gross oxygen productivity coming from the terrestrial or marine biosphere on $\Delta'^{17}O\ O_{2,\ trop}$. The Rice University model run and the Wostbrock et al. (2020) observation are in shades of blue.[6] The Hebrew University model run and the Luz and Barkan (2011) observation are in shades of red.[32] No relative proportion of terrestrial versus marine gross oxygen productivity brings model results in line with observations in either reference frame.

Figure 7



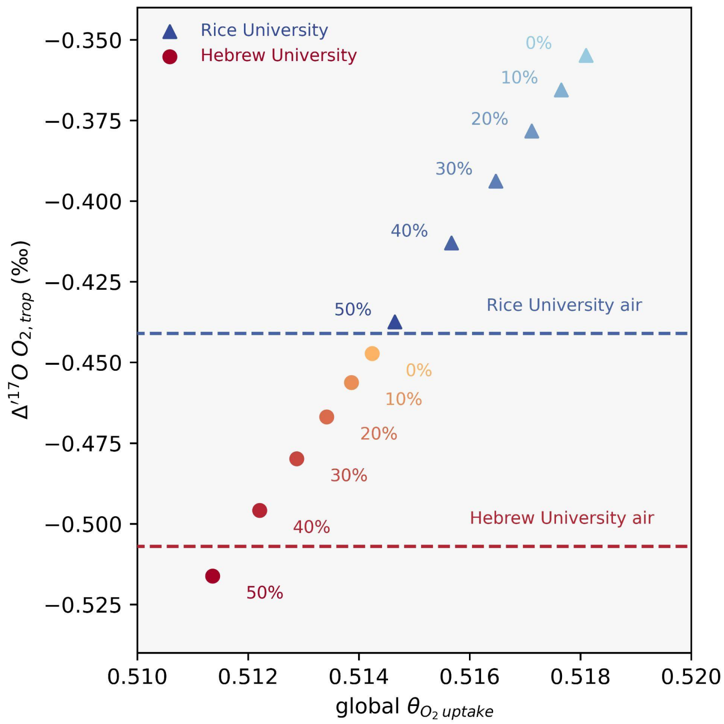
Caption: Plot showing the effect of changing the value of the global $\theta_{O2\ uptake}$ to the $\theta_{O2\ uptake}$ characterizing different biological O_2 consumption mechanisms on $\Delta'^{17}O\ O_{2,\ trop}$. Citations for the plotted $\theta_{O2\ uptake}$ values can be found in the text. The Rice University model run and Wostbrock et al. (2020) observation are in shades of blue.[6] The Hebrew University model run and Luz and Barkan (2011) observation are in shades of red.[32] MRt - terrestrial Mehler reaction, DR - dark respiration, COX - cytochrome c oxidase pathway, AOX - alternative oxidase pathway, PR - photorespiration, MRm - marine Mehler reaction.

Figure 8



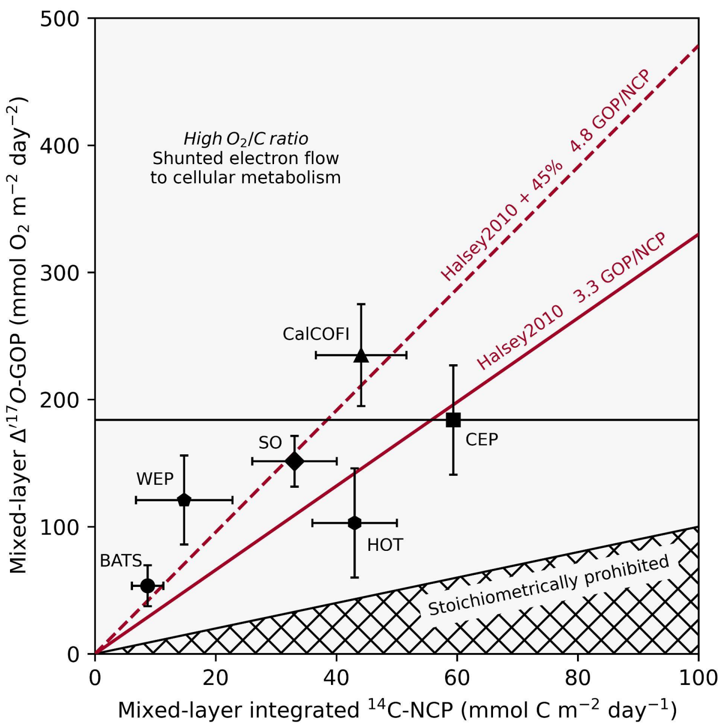
Caption: Plot showing the effect of changing the fraction of $\theta_{O2\ uptake}$ coming from the cytochrome c oxidase pathway or photorespiration on $\Delta'^{17}O\ O_{2,\ trop}$. The Rice University model run and the Wostbrock et al. (2020) observation are in shades of blue.[6] The Hebrew University model run and the Luz and Barkan (2011) observation are in shades of red.[32] 90% of global biological O_2 consumption occurring via photorespiration brings the Rice University model result in line the corresponding air observation. A 50/50 global breakdown of the cytochrome c oxidase pathway versus photorespiration brings the Hebrew University model results in line with the corresponding air observation.

Figure 9



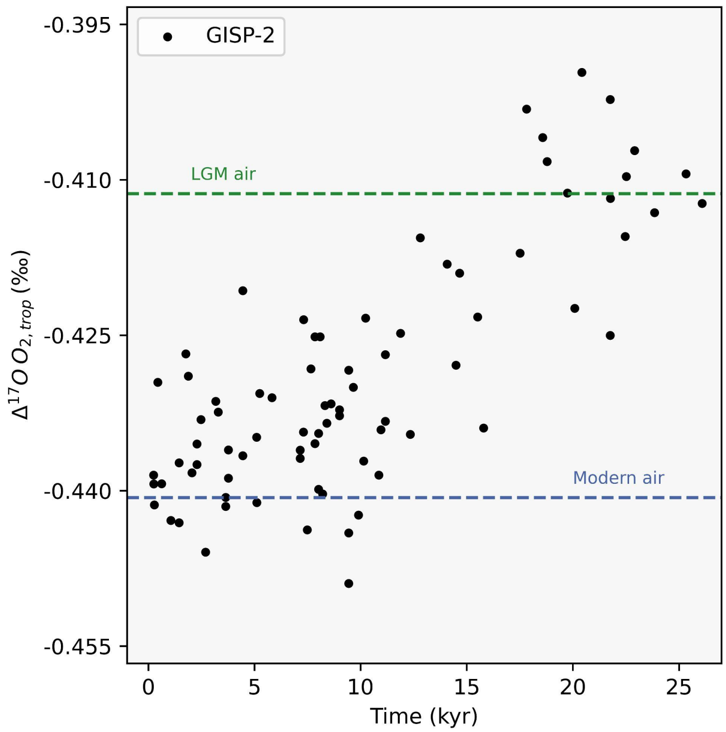
Caption: Plot showing the predicted $\Delta'^{17}O$ $O_{2,\,trop}$ for a range of Mehler-like reactions O_2 uptake fluxes spanning 0 to 50% of the total marine O_2 uptake. Rice University model results and the Wostbrock et al. (2020) observation are in shades of blue.[6] Hebrew University model results and the Luz and Barkan (2011) observation are in shades of red.[32] Observations are in-line with model outputs for a Mehler-like reaction O_2 flux equal to between 40% and 50% of total marine O_2 uptake in both reference frames.

Figure 10



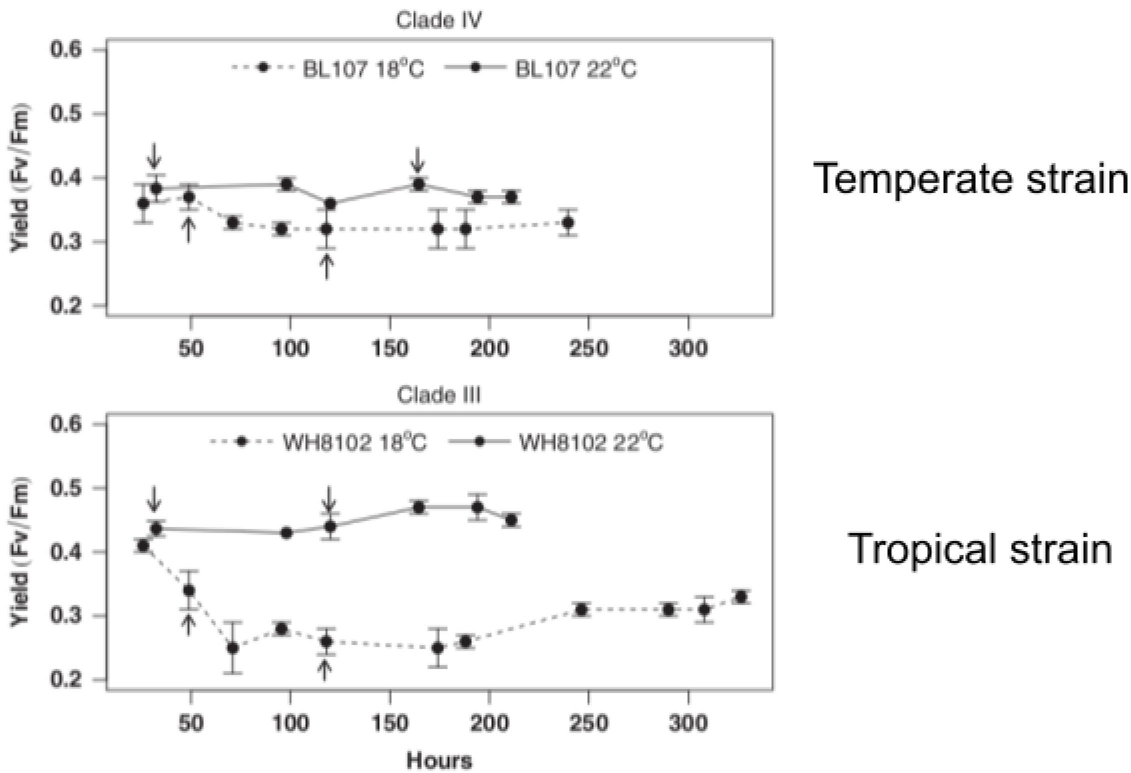
Caption: Plot showing $\Delta'^{17}O$-based gross oxygen production versus ^{14}C-based net carbon fixation data. Citations for data plotted can be found in the text. Uncertainties are one σ. The gross oxygen productivity to net carbon productivity ratio of 3.3 comes from culture experiments conducted in Halsey et al. (2010).[37] The gross oxygen productivity to net carbon productivity ratio 4.8 was calculated by multiplying 3.3 by 1.45 to obtain a slope including the effect of a Mehler-like reaction O_2 flux equal to 45% of gross oxygen productivity. The higher slope explains more of the data than the lower slope does.

Figure 11



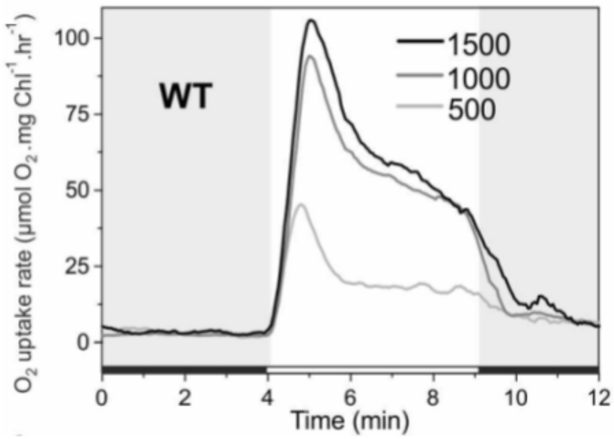
Caption: Plot showing the variation in $\Delta'^{17}O$ of tropospheric O_2 extracted glacial ice collected from the GISP-2 drilling site since the Last Glacial Maximum as reported in Blunier et al. (2012).[2]

Figure 12



Caption: Comparison between the relative quantities of O_2 production consumed by Mehler-like reactions in experiments run at 18 °C and 22 °C on temperate and tropical strains of *Synechococccus* from Varkey et al. (2016).[76]

Figure 13



Strain	Light intensity (µmol photons. $m^{-2}.s^{-1}$)	Maximum rate of light-induced O_2 uptake (O_2 µmol.mgChl^{-1}.hr^{-1})
WT	500	41.2 ± 6.0
	1000	87.7 ± 7.2
	1500	96.4 ± 9.1

Caption: Plot and table showing the increase in the flux of O_2 consumed by marine Mehler-like reactions at higher light intensities found in Santana-Sanchez et al. (2019).[21]

Tables

Table 1

Species	**Box**
O	Stratosphere
^{17}O	Stratosphere
^{18}O	Stratosphere
$O(^1D)$	Stratosphere
$^{17}O(^1D)$	Stratosphere
$^{18}O(^1D)$	Stratosphere
O_2	Stratosphere
$O^{17}O$	Stratosphere
$O^{18}O$	Stratosphere
CO_2	Stratosphere
$CO^{17}O$	Stratosphere
$CO^{18}O$	Stratosphere
O_3	Stratosphere
$OO^{17}O$	Stratosphere

$OO^{18}O$	Stratosphere
O_2	Troposphere
$O^{17}O$	Troposphere
$O^{18}O$	Troposphere
CO_2	Troposphere
$CO^{17}O$	Troposphere
$CO^{18}O$	Troposphere
O_2	Biosphere
$O^{17}O$	Biosphere
$O^{18}O$	Biosphere

Table 2

Reaction	**Reaction Name**	**Reaction Rate from Literature**
$O_2 + hv \rightarrow O + O$	R1A	1.109×10^{-12} (1 / s)
$O^{18}O + hv \rightarrow O + {}^{18}O$	R1B	R1A
$O^{17}O + hv \rightarrow O + {}^{17}O$	R1C	R1A
$O_2 + O + M \rightarrow O_3$	R2A	$1.26 \times 10^{-33} \times 8.3 \times 10^{17}$ cm^3 second^{-1}
$O_2 + {}^{18}O + M \rightarrow OO^{18}O$	R2B	R2A × α_{MIF}
$O_2 + {}^{17}O + M \rightarrow OO^{17}O$	R2C	R2A × α_{MIF}
$O^{18}O + O + M \rightarrow OO^{18}O$	R2D	R2A × α_{MIF}
$O^{17}O + O + M \rightarrow OO^{17}O$	R2E	R2A × α_{MIF}
$O_3 + hv \rightarrow O_2 + O$	R3A	2.96×10^{-4} (1 / s)
$OO^{18}O + hv \rightarrow O_2 + {}^{18}O$	R3B	R3A × (1/3)
$OO^{18}O + hv \rightarrow O^{18}O + O$	R3C	R3A × (2/3)
$OO^{17}O + hv \rightarrow O_2 + {}^{17}O$	R3D	R3A × (1/3)
$OO^{17}O + hv \rightarrow O^{17}O + O$	R3E	R3A × (2/3)
$O_3 + hv \rightarrow O_2 + O({}^1D)$	R4A	5.01×10^{-4} second^{-1}

$OO^{18}O + h\nu \rightarrow O_2 + {}^{18}O(^1D)$	R4B	R4A × (1/3)
$OO^{18}O + h\nu \rightarrow O^{18}O + O(^1D)$	R4C	R4A × (2/3)
$OO^{17}O + h\nu \rightarrow O_2 + {}^{17}O(^1D)$	R4D	R4A × (1/3)
$OO^{17}O + h\nu \rightarrow O^{17}O + O(^1D)$	R4E	R4A × (2/3)
$O_3 + O \rightarrow O_2 + O_2$	R5A	6.86×10^{-16} cm^3 $second^{-1}$
$OO^{18}O + O \rightarrow O_2 + O^{18}O$	R5B	R5A
$OO^{17}O + O \rightarrow O_2 + O^{17}O$	R5C	R5A
$O_3 + {}^{18}O \rightarrow O_2 + O^{18}O$	R5D	R5A
$O_3 + {}^{17}O \rightarrow O_2 + O^{17}O$	R5E	R5A
$O_3 + O(^1D) \rightarrow O_2 + O_2$	R6A	2.4×10^{-10} cm^3 $second^{-1}$
$OO^{18}O + O(^1D) \rightarrow O_2 + O^{18}O$	R6B	R6A × (1/2)
$OO^{17}O + O(^1D) \rightarrow O_2 + O^{17}O$	R6C	R6A × (1/2)
$O_3 + {}^{18}O(^1D) \rightarrow O_2 + O^{18}O$	R6D	R6A × (1/2)
$O_3 + {}^{17}O(^1D) \rightarrow O_2 + O^{17}O$	R6E	R6A × (1/2)
$O_3 + O(^1D) \rightarrow O_2 + O + O$	R6F	R6A × (1/2)
$OO^{18}O + O(^1D) \rightarrow O_2 + O + {}^{18}O$	R6G	R6A × (1/2) × (1/2)

$OO^{18}O + O(^1D) \rightarrow O^{18}O + O + O$	R6H	R6A × (1/2) × (1/2)
$OO^{17}O + O(^1D) \rightarrow O_2 + O + {}^{17}O$	R6I	R6A × (1/2) × (1/2)
$OO^{17}O + O(^1D) \rightarrow O^{17}O + O + O$	R6J	R6A × (1/2) × (1/2)
$O_3 + {}^{18}O(^1D) \rightarrow O_2 + O + {}^{18}O$	R6K	R6A × (1/2) × (1/2)
$O_3 + {}^{18}O(^1D) \rightarrow O^{18}O + O + O$	R6L	R6A × (1/2) × (1/2)
$O_3 + {}^{17}O(^1D) \rightarrow O_2 + O + {}^{17}O$	R6M	R6A × (1/2) x (1/2)
$O_3 + {}^{17}O(^1D) \rightarrow O^{17}O + O + O$	R6N	R6A × (1/2) × (1/2)
$O_2 + O(^1D) \rightarrow O_2 + O$	R7A	$4.24 \times 10^{-11} \times (0.78 + 0.21) / 0.21$ cm^3 second^{-1}
$O_2 + {}^{18}O(^1D) \rightarrow O_2 + {}^{18}O$	R7B	R7A
$O_2 + {}^{18}O(^1D) \rightarrow O^{18}O + O$	R7C	R7A
$O_2 + {}^{17}O(^1D) \rightarrow O_2 + {}^{17}O$	R7D	R7A
$O_2 + O_2 + {}^{17}O(^1D) \rightarrow O^{17}O + O$	R7E	R7A
$O_2 + {}^{18}O \rightarrow O^{18}O + O$	R8A	2.42×10^{-6} cm^3 second^{-1}
$O^{18}O + O \rightarrow O_2 + {}^{18}O$	R8B	R8A × (1/2)
$O_2 + {}^{17}O \rightarrow O^{17}O + O$	R8C	R8A

$O^{17}O + O \rightarrow O_2 + {}^{17}O$	R8D	R8A × (1/2)
$CO_2 + O(^1D) \rightarrow CO_2 + O$	R9A	1.26×10^{-10} cm^3 $second^{-1}$
$CO_2 + {}^{18}O(^1D) \rightarrow CO_2 + {}^{18}O$	R9B	R9A × (1/3)
$CO_2 + {}^{18}O(^1D) \rightarrow CO_2 + {}^{18}O$	R9C	R9A × (2/3)
$CO^{18}O + O(^1D) \rightarrow CO_2 + {}^{18}O$	R9D	R9A × (1/3)
$CO^{18}O + O(^1D) \rightarrow CO^{18}O + O$	R9E	R9A × (2/3)
$CO_2 + {}^{17}O(^1D) \rightarrow CO_2 + {}^{17}O$	R9F	R9A × (1/3)
$CO_2 + {}^{17}O(^1D) \rightarrow CO^{17}O + O$	R9G	R9A × (2/3)
$CO^{17}O + O(^1D) \rightarrow CO_2 + {}^{17}O$	R9H	R9A × (1/3)
$CO^{17}O + O(^1D) \rightarrow CO^{17}O + O$	R9I	R9A × (2/3)

Table 3

Isotopic Composition	**Target Value (‰)**	**Model Result (‰)**
$\delta'^{17}O$ O_2 Troposphere	12.105 ± 0.065	12.155
$\delta'^{18}O$ O_2 Troposphere	23.761 ± 0.104	23.687
$\Delta'^{17}O$ O_2 Troposphere	-0.441 ± 0.012	-0.337
$\delta'^{17}O$ CO_2 Stratosphere	22.9	25.6
$\delta'^{18}O$ CO_2 Stratosphere	40.4	45.6
$\Delta'^{17}O$ CO_2 Stratosphere	1.55	1.54
$\delta'^{17}O$ CO_2 Troposphere	21.3	22.5
$\delta'^{18}O$ CO_2 Troposphere	40.0	42.5
$\Delta'^{17}O$ CO_2 Troposphere	0.23	0.05
$\delta'^{17}O$ O_3 Stratosphere	77 ± 10	75
$\delta'^{18}O$ O_3 Stratosphere	94 ± 13	87
$\Delta'^{17}O$ O_3 Stratosphere	29 ± 4	29
Mole Fraction	**Target Value**	**Model Result**
O_2 Troposphere	0.21	0.21

CO_2 Troposphere	2.7×10^{-4}	2.7×10^{-4}
CO_2 Stratosphere	2.7×10^{-4}	2.7×10^{-4}
O_3 Stratosphere	2×10^{-6}	6.6×10^{-6}
Isotopic Flux	**Target Value**	**Model Result**
$\Delta'^{17}O$ CO_2 Stratosphere - Troposphere	4×10^{15} ‰ mole year^{-1}	7.4×10^{15} ‰ mole year^{-1}

Table 4

Theta Type	**Citation**	**Hebrew University**	**Rice University**
θ_{COX}	Angert et al. (2003)[15]	0.516 ± 0.001	
$\theta_{dark\ resp.}$	Ash et al. (2020)[9]		0.520 ± 0.001
θ_{PR}	Helman et al. (2005)[28]	0.509 ± 0.002	0.513 ± 0.002
θ_{AOX}	Angert et al. (2003)	0.514 ± 0.001	0.518 ± 0.001
θ_{MRt}	Helman et al. (2005)	0.525 ± 0.002	0.525 ± 0.002
θ_{MRm}	Helman et al. (2005)	0.496 ± 0.004	0.496 ± 0.004

Bibliography

1. Juranek, L. W. & Quay, P. D. Using Triple Isotopes of Dissolved Oxygen to Evaluate Global Marine Productivity. *Annu. Rev. Mar. Sci.* **5**, 503–524 (2013).
2. Blunier, T., Bender, M. L., Barnett, B. & von Fischer, J. C. Planetary fertility during the past 400 ka based on the triple isotope composition of O2 in trapped gases from the Vostok ice core. *Clim. Past* **8**, 1509–1526 (2012).
3. Young, E. D., Yeung, L. Y. & Kohl, I. E. On the Δ17O budget of atmospheric O2. *Geochim. Cosmochim. Acta* **135**, 102–125 (2014).
4. Yeung, L. Y., Hayles, J. A., Hu, H., Ash, J. L. & Sun, T. Scale distortion from pressure baselines as a source of inaccuracy in triple-isotope measurements. *Rapid Commun. Mass Spectrom.* **32**, 1811–1821 (2018).
5. Pack, A. *et al.* Tracing the oxygen isotope composition of the upper Earth's atmosphere using cosmic spherules. *Nat. Commun.* **8**, 15702 (2017).
6. Wostbrock, J. A. G., Cano, E. J. & Sharp, Z. D. An internally consistent triple oxygen isotope calibration of standards for silicates, carbonates and air relative to VSMOW2 and SLAP2. *Chem. Geol.* **533**, 119432 (2020).
7. Barkan, E. & Luz, B. The relationships among the three stable isotopes of oxygen in air, seawater and marine photosynthesis. *Rapid Commun. Mass Spectrom. RCM* **25**, 2367–2369 (2011).
8. Luz, B. & Barkan, E. Net and gross oxygen production from O2/Ar, 17O/16O and 18O/16O ratios. *Aquat. Microb. Ecol.* **56**, 133–145 (2009).
9. Ash, J. L., Hu, H. & Yeung, L. Y. What Fractionates Oxygen Isotopes during Respiration? Insights from Multiple Isotopologue Measurements and Theory. *ACS Earth Space Chem.* **4**,

50–66 (2020).

10. Yung, Y. L., DeMore, W. B. & Pinto, J. P. Isotopic exchange between carbon dioxide and ozone via O(1D) in the stratosphere. *Geophys. Res. Lett.* **18**, 13–16 (1991).

11. Young, E. D., Galy, A. & Nagahara, H. Kinetic and equilibrium mass-dependent isotope fractionation laws in nature and their geochemical and cosmochemical significance. *Geochim. Cosmochim. Acta* **66**, 1095–1104 (2002).

12. Capaldi, R. A. STRUCTURE AND FUNCTION OF CYTOCHROME C OXIDASE. *Annu. Rev. Biochem.* **59**, 569–596 (1990).

13. Blunier, T., Barnett, B., Bender, M. L. & Hendricks, M. B. Biological oxygen productivity during the last 60,000 years from triple oxygen isotope measurements. *Glob. Biogeochem. Cycles* **16**, 3-1-3–13 (2002).

14. Vanlerberghe, G. C. & McIntosh, L. ALTERNATIVE OXIDASE: From Gene to Function. *Annu. Rev. Plant Physiol. Plant Mol. Biol.* **48**, 703–734 (1997).

15. Angert, A., Rachmilevitch, S., Barkan, E. & Luz, B. Effects of photorespiration, the cytochrome pathway, and the alternative pathway on the triple isotopic composition of atmospheric O2. *Glob. Biogeochem. Cycles* **17**, (2003).

16. Dellero, Y., Jossier, M., Schmitz, J., Maurino, V. G. & Hodges, M. Photorespiratory glycolate–glyoxylate metabolism. *J. Exp. Bot.* **67**, 3041–3052 (2016).

17. Sharkey, T. D. Estimating the rate of photorespiration in leaves. *Physiol. Plant.* **73**, 147–152 (1988).

18. Bauwe, H., Hagemann, M. & Fernie, A. R. Photorespiration: players, partners and origin. *Trends Plant Sci.* **15**, 330–336 (2010).

19. Moroney, J. V., Jungnick, N., DiMario, R. J. & Longstreth, D. J. Photorespiration and carbon

concentrating mechanisms: two adaptations to high O2, low CO2 conditions. *Photosynth. Res.* **117**, 121–131 (2013).

20. Osmond, C. B. *et al.* Photorespiration: metabolic pathways and their role in stress protection. *Philos. Trans. R. Soc. Lond. B. Biol. Sci.* **355**, 1517–1529 (2000).

21. Santana-Sanchez, A. *et al.* Flavodiiron proteins 1–to-4 function in versatile combinations in O2 photoreduction in cyanobacteria. *eLife* **8**, e45766 (2019).

22. Allahverdiyeva, Y., Suorsa, M., Tikkanen, M. & Aro, E.-M. Photoprotection of photosystems in fluctuating light intensities. *J. Exp. Bot.* **66**, 2427–2436 (2015).

23. Asada, K. The water–water cycle as alternative photon and electron sinks. *Philos. Trans. R. Soc. Lond. B. Biol. Sci.* **355**, 1419–1431 (2000).

24. Asada, K. THE WATER-WATER CYCLE IN CHLOROPLASTS: Scavenging of Active Oxygens and Dissipation of Excess Photons. *Annu. Rev. Plant Physiol. Plant Mol. Biol.* **50**, 601–639 (1999).

25. Badger, M. R., von Caemmerer, S., Ruuska, S. & Nakano, H. Electron flow to oxygen in higher plants and algae: rates and control of direct photoreduction (Mehler reaction) and rubisco oxygenase. *Philos. Trans. R. Soc. B Biol. Sci.* **355**, 1433–1446 (2000).

26. Allahverdiyeva, Y. *et al.* Flavodiiron proteins Flv1 and Flv3 enable cyanobacterial growth and photosynthesis under fluctuating light. *Proc. Natl. Acad. Sci.* **110**, 4111–4116 (2013).

27. Goosney, D. L. & Miller, A. G. High rates of O2 photoreduction by the unicellular cyanobacterium Synechocystis PCC 6803 as determined by the quenching of chlorophyll fluorescence. *Can. J. Bot.* **75**, 394–401 (1997).

28. Helman, Y., Barkan, E., Eisenstadt, D., Luz, B. & Kaplan, A. Fractionation of the Three Stable Oxygen Isotopes by Oxygen-Producing and Oxygen-Consuming Reactions in

Photosynthetic Organisms. *Plant Physiol.* **138**, 2292–2298 (2005).

29. Roberty, S., Bailleul, B., Berne, N., Franck, F. & Cardol, P. PSI Mehler reaction is the main alternative photosynthetic electron pathway in Symbiodinium sp., symbiotic dinoflagellates of cnidarians. *New Phytol.* **204**, 81–91 (2014).

30. Allahverdiyeva, Y. *et al.* Interplay between Flavodiiron Proteins and Photorespiration in Synechocystis sp. PCC 6803*. *J. Biol. Chem.* **286**, 24007–24014 (2011).

31. Theune, M. L. *et al.* In-vivo quantification of electron flow through photosystem I – Cyclic electron transport makes up about 35% in a cyanobacterium. *Biochim. Biophys. Acta BBA - Bioenerg.* **1862**, 148353 (2021).

32. Luz, B. & Barkan, E. Proper estimation of marine gross O2 production with 17O/16O and 18O/16O ratios of dissolved O2. *Geophys. Res. Lett.* **38**, (2011).

33. Yeung, L. Y., Hayles, J. A., Hu, H., Ash, J. L. & Sun, T. Scale distortion from pressure baselines as a source of inaccuracy in triple-isotope measurements. *Rapid Commun. Mass Spectrom.* **32**, 1811–1821 (2018).

34. Luz, B. & Barkan, E. The isotopic ratios 17O/16O and 18O/16O in molecular oxygen and their significance in biogeochemistry. *Geochim. Cosmochim. Acta* **69**, 1099–1110 (2005).

35. Stolper, D. A., Fischer, W. W. & Bender, M. L. Effects of temperature and carbon source on the isotopic fractionations associated with O2 respiration for 17O/16O and 18O/16O ratios in E. coli. *Geochim. Cosmochim. Acta* **240**, 152–172 (2018).

36. Regaudie-de-Gioux, A., Lasternas, S., Agustí, S. & Duarte, C. M. Comparing marine primary production estimates through different methods and development of conversion equations. *Front. Mar. Sci.* **1**, (2014).

37. Halsey, K. H., Milligan, A. J. & Behrenfeld, M. J. Physiological optimization underlies

growth rate-independent chlorophyll-specific gross and net primary production. *Photosynth. Res.* **103**, 125–137 (2010).

38. Burkholder, J. B. *et al. Chemical kinetics and photochemical data for use in atmospheric studies; evaluation number 19*. https://trs.jpl.nasa.gov/handle/2014/49199 (2020).

39. Sander, S. P. *et al. Chemical kinetics and photochemical data for use in Atmospheric Studies Evaluation Number 15*. https://trs.jpl.nasa.gov/handle/2014/41648 (2006).

40. DeMore, W. B., Sander, S. P., Howard, C. J. & Ravishankara, A. R. JPL Publication 15-10: Chemical Kinetics and Photochemical Data for Use in Atmospheric Studies. 278 (1997).

41. Matsumi, Y. & Kawasaki, M. Photolysis of Atmospheric Ozone in the Ultraviolet Region. *Chem. Rev.* **103**, 4767–4782 (2003).

42. Fleurat-Lessard, P., Grebenshchikov, S. Yu., Schinke, R., Janssen, C. & Krankowsky, D. Isotope dependence of the O+O2 exchange reaction: Experiment and theory. *J. Chem. Phys.* **119**, 4700–4712 (2003).

43. Beck, W. C., Grossman, E. L. & Morse, J. W. Experimental studies of oxygen isotope fractionation in the carbonic acid system at 15°, 25°, and 40°C. *Geochim. Cosmochim. Acta* **69**, 3493–3503 (2005).

44. Welp, L. R. *et al.* Interannual variability in the oxygen isotopes of atmospheric CO 2 driven by El Niño. *Nature* **477**, 579–582 (2011).

45. West, J. B., Sobek, A. & Ehleringer, J. R. A Simplified GIS Approach to Modeling Global Leaf Water Isoscapes. *PLOS ONE* **3**, e2447 (2008).

46. Farquhar, G. D. *et al.* Vegetation effects on the isotope composition of oxygen in atmospheric CO2. *Nature* **363**, 439–443 (1993).

47. Landais, A., Barkan, E., Yakir, D. & Luz, B. The triple isotopic composition of oxygen in

leaf water. *Geochim. Cosmochim. Acta* **70**, 4105–4115 (2006).

48. Held, I. M. & Soden, B. J. Robust Responses of the Hydrological Cycle to Global Warming. *J. Clim.* **19**, 5686–5699 (2006).

49. Wiegel, A. A. *et al.* Unexpected variations in the triple oxygen isotope composition of stratospheric carbon dioxide. *Proc. Natl. Acad. Sci.* **110**, 17680–17685 (2013).

50. Appenzeller, C., Holton, J. R. & Rosenlof, K. H. Seasonal variation of mass transport across the tropopause. *J. Geophys. Res. Atmospheres* **101**, 15071–15078 (1996).

51. Trenberth, K. E. & Smith, L. The Mass of the Atmosphere: A Constraint on Global Analyses. *J. Clim.* **18**, 864–875 (2005).

52. Liang, J. 1 - Chemical composition of the atmosphere of the Earth. in *Chemical Modeling for Air Resources* 3–20 (Academic Press, 2013). doi:10.1016/B978-0-12-408135-2.00001-X.

53. Bender, M., Sowers, T. & Labeyrie, L. The Dole Effect and its variations during the last 130,000 years as measured in the Vostok Ice Core. *Glob. Biogeochem. Cycles* **8**, 363–376 (1994).

54. Luz, B., Barkan, E. & Severinghaus, J. P. The Stable Isotopic Composition of Atmospheric O2. in *Treatise on Geochemistry* 363–383 (Elsevier, 2014). doi:10.1016/B978-0-08-095975-7.00419-8.

55. Eisenstadt, D., Barkan, E., Luz, B. & Kaplan, A. Enrichment of oxygen heavy isotopes during photosynthesis in phytoplankton. *Photosynth. Res.* **103**, 97–103 (2010).

56. Tolbert, N. E. The C2 Oxidative Photosynthetic Carbon Cycle. *Annu. Rev. Plant Physiol. Plant Mol. Biol.* **48**, 1–25 (1997).

57. Wigley, T. M. L. The pre-industrial carbon dioxide level. *Clim. Change* **5**, 315–320 (1983).

58. Krankowsky, D. *et al.* Stratospheric ozone isotope fractionations derived from collected

samples. *J. Geophys. Res. Atmospheres* **112**, (2007).

59. Boering, K. A. *et al.* Observations of the anomalous oxygen isotopic composition of carbon dioxide in the lower stratosphere and the flux of the anomaly to the troposphere. *Geophys. Res. Lett.* **31**, (2004).

60. Liang, M.-C., Mahata, S., Laskar, A. H., Thiemens, M. H. & Newman, S. Oxygen isotope anomaly in tropospheric CO 2 and implications for CO 2 residence time in the atmosphere and gross primary productivity. *Sci. Rep.* **7**, 13180 (2017).

61. Ometto, J. P. H., Flanagan, L. B., Martinelli, L. A. & Ehleringer, J. R. Oxygen Isotope Ratios of Waters and Respired CO2 in Amazonian Forest and Pasture Ecosystems. *Ecol. Appl.* **15**, 58–70 (2005).

62. Tans, P. P. Oxygen isotopic equilibrium between carbon dioxide and water in soils. *Tellus B* **50**, 163–178 (1998).

63. Zhang, C., Dazlich, D. A., Randall, D. A., Sellers, P. J. & Denning, A. S. Calculation of the global land surface energy, water and CO2 fluxes with an off-line version of SiB2. *J. Geophys. Res. Atmospheres* **101**, 19061–19075 (1996).

64. François, L. M., Delire, C., Warnant, P. & Munhoven, G. Modelling the glacial–interglacial changes in the continental biosphere. *Glob. Planet. Change* **16–17**, 37–52 (1998).

65. Keeling, R. F. Development of an interferometric oxygen analyzer for precise measurement of the atmospheric O2 mole fraction. *Ph Thesis Harv. Univ.* (1988).

66. von Caemmerer, S. & Farquhar, G. D. Some relationships between the biochemistry of photosynthesis and the gas exchange of leaves. *Planta* **153**, 376–387 (1981).

67. Field, C. B., Behrenfeld, M. J., Randerson, J. T. & Falkowski, P. Primary Production of the Biosphere: Integrating Terrestrial and Oceanic Components. *Science* **281**, 237–240 (1998).

68. Stanley, R. H. R., Kirkpatrick, J. B., Cassar, N., Barnett, B. A. & Bender, M. L. Net community production and gross primary production rates in the western equatorial Pacific. *Glob. Biogeochem. Cycles* **24**, (2010).

69. Hamme, R. C. *et al.* Dissolved O2/Ar and other methods reveal rapid changes in productivity during a Lagrangian experiment in the Southern Ocean. *J. Geophys. Res. Oceans* **117**, (2012).

70. Luz, B. & Barkan, E. Assessment of Oceanic Productivity with the Triple-Isotope Composition of Dissolved Oxygen. *Science* **288**, 2028–2031 (2000).

71. Quay, P. D., Peacock, C., Björkman, K. & Karl, D. M. Measuring primary production rates in the ocean: Enigmatic results between incubation and non-incubation methods at Station ALOHA. *Glob. Biogeochem. Cycles* **24**, n/a-n/a (2010).

72. Munro, D. R., Quay, P. D., Juranek, L. W. & Goericke, R. Biological production rates off the Southern California coast estimated from triple O2 isotopes and O2 : Ar gas ratios. *Limnol. Oceanogr.* **58**, 1312–1328 (2013).

73. Fujiki, T., Suzue, T., Kimoto, H. & Saino, T. Photosynthetic electron transport in Dunaliella tertiolecta (Chlorophyceae) measured by fast repetition rate fluorometry: relation to carbon assimilation. *J. Plankton Res.* **29**, 199–208 (2007).

74. Tierney, J. E. *et al.* Glacial cooling and climate sensitivity revisited. *Nature* **584**, 569–573 (2020).

75. Flombaum, P. *et al.* Present and future global distributions of the marine Cyanobacteria Prochlorococcus and Synechococcus. *Proc. Natl. Acad. Sci.* **110**, 9824–9829 (2013).

76. Varkey, D. *et al.* Effects of low temperature on tropical and temperate isolates of marine Synechococcus. *ISME J.* **10**, 1252–1263 (2016).

77. Clark, P. U. *et al.* The Last Glacial Maximum. *Science* **325**, 710–714 (2009).

78. Behrenfeld, M. J. Widespread Iron Limitation of Phytoplankton in the South Pacific Ocean. *Science* **283**, 840–843 (1999).

79. Allahverdiyeva, Y., Isojärvi, J., Zhang, P. & Aro, E.-M. Cyanobacterial Oxygenic Photosynthesis is Protected by Flavodiiron Proteins. *Life* **5**, 716–743 (2015).

80. Albani, S. *et al.* Paleodust variability since the Last Glacial Maximum and implications for iron inputs to the ocean. *Geophys. Res. Lett.* **43**, 3944–3954 (2016).

81. Guy, R. D., Fogel, M. L. & Berry, J. A. Photosynthetic Fractionation of the Stable Isotopes of Oxygen and Carbon. *Plant Physiol.* **101**, 37–47 (1993).

82. Yeung, L. Y., Ash, J. L. & Young, E. D. Biological signatures in clumped isotopes of O2. *Science* **348**, 431–434 (2015).

83. Kanno, M., Carroll, A. L. & Atsumi, S. Global metabolic rewiring for improved CO2 fixation and chemical production in cyanobacteria. *Nat. Commun.* **8**, 14724 (2017).

84. Shultzaberger, R. K., Boyd, J. S., Diamond, S., Greenspan, R. J. & Golden, S. S. Giving Time Purpose: The Synechococcus elongatus Clock in a Broader Network Context. *Annu. Rev. Genet.* **49**, 485–505 (2015).

85. Helman, Y. *et al.* Genes Encoding A-Type Flavoproteins Are Essential for Photoreduction of O2 in Cyanobacteria. *Curr. Biol.* **13**, 230–235 (2003).

86. Kaplan, A. & Reinhold, L. CO2 Concentrating Mechanisms in Photosynthetic Microorganisms. *Annu. Rev. Plant Physiol. Plant Mol. Biol.* **50**, 539–570 (1999).

87. Mustila, H. *et al.* The Flavodiiron Protein Flv3 Functions as a Homo-Oligomer During Stress Acclimation and is Distinct from the Flv1/Flv3 Hetero-Oligomer Specific to the O2 Photoreduction Pathway. *Plant Cell Physiol.* **57**, 1468–1483 (2016).